

Match Quality and Firm Dynamism

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Abstract

Match quality plays a central role in theories of job creation, separation, and worker mobility. Yet it is rarely measured directly, limiting our understanding of its broader economic implications. This paper develops a new measure of match quality using a matched employer and employee panel constructed from LinkedIn profiles provided by Revelio Labs. We estimate a multichannel latent Markov model that classifies employment spells into seven latent match states based on their duration and exit outcomes. These outcomes include movements across firms and changes within a firm's internal hierarchy. The recovered states range from short, separation prone positions to stable positions with long durations and opportunities for internal advancement. Entry wages, which are excluded from the estimation, broadly increase across this ranking. We use posterior state probabilities to examine how match states vary with worker and firm characteristics. Younger and less educated workers are more likely to occupy separation prone states, while larger and older firms are more likely to form matches associated with stability and internal growth. Finally, we construct match innovations by comparing each position's realized state distribution with the distribution predicted from the worker's characteristics and prior employment history. These innovations isolate the unexpected component of match quality from systematic worker and firm heterogeneity. Firms experiencing more favorable match innovations subsequently grow faster, have lower separation rates, and are more likely to survive. These relationships remain substantial when identified using variation within the same firm over time.

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1 Introduction

In many macro labor models, a time varying match specific shock governs job creation and separation. These models often feature a matching function with an idiosyncratic productivity draw that determines whether a job is created and when it is dissolved. Match quality also plays a central role in models of on the job search and job ladders, where workers move through the wage distribution as they receive better or worse match realizations. Despite its prominent theoretical role, relatively few empirical studies directly measure match quality or examine its connection to firm dynamism. This paper fills that gap. We ask whether firms whose workers experience favorable match realizations subsequently grow faster, retain more employees, and survive at higher rates.

We answer this question using a novel matched employer and employee panel from Revelio Labs that is constructed from individual LinkedIn profiles. Our empirical analysis proceeds in three steps. First, we estimate a latent Markov model that identifies distinct latent match states from the observed trajectory of each employment relationship. Second, we construct a position level match innovation that captures the component of the realized match state that could not have been predicted from the worker’s characteristics and employment history before the match. Third, we aggregate these innovations to the firm level and examine their relationship with subsequent employment growth, separation rates, and survival. We find that match innovations have substantial effects on subsequent firm outcomes, even after controlling for firm fixed effects.

A key advantage of the Revelio data is that they track workers’ employment histories at the position level. We observe not only transitions across firms, but also promotions, lateral transfers, and downgrades within firms, together with the timing of each change. Administrative wage records typically identify worker and firm relationships but do not reveal workers’ movement through firms’ internal hierarchies. The Revelio data therefore allow us to characterize the evolution of a match along a seniority dimension that is distinct from wages.

In the first step, we estimate a multichannel latent Markov model that classifies each employment spell into one of seven latent match states. The classification is based on the joint distribution of spell duration and exit type. Exit types include internal promotions and transfers, movements to other firms, transitions into education or nonemployment, relocation, and ongoing employment. Each state is characterized by its emission probabilities, which describe the distribution of observed durations and exit outcomes associated with that state, and by its transition probabilities, which describe how workers move between latent states across successive employment spells. Because the model does not use

worker demographics, firm characteristics, or wages to define the states, it provides a fixed classification based only on the observed evolution of employment relationships.

The seven states range from short, separation prone positions to stable jobs with long durations and opportunities for internal advancement. We assess the economic content of this classification using wages, which are excluded from the estimation. Entry wages generally increase as the states move from temporary and separation prone positions toward stable positions with internal growth. This relationship provides an external check that the latent states capture economically meaningful differences in match quality.

In the second step, we use soft decoding to assign each position a posterior probability of belonging to each of the seven states. Rather than assigning a position to a single state, soft decoding retains the entire posterior distribution and therefore accounts for uncertainty in the classification. We then estimate spell level regressions relating these posterior probabilities to worker and firm characteristics. Younger and less educated workers are more likely to hold separation prone positions, while larger and older firms are more likely to provide positions associated with stability and internal growth.

In the third step, we re-estimate the initial state and transition equations using a richer set of predetermined worker characteristics. These equations generate a predicted distribution across match states for each position based on information available before the match, including the worker's demographic characteristics and prior employment history. We compare this predicted distribution with the position's realized posterior state distribution. The difference between the two is our measure of the match-specific innovation. It captures whether the realized employment relationship was unexpectedly favorable or unfavorable relative to what could have been predicted when the match began.

This innovation provides an empirical counterpart to the idiosyncratic match-quality shock that governs job formation and separation in search models with match-specific productivity, such as Jovanovic (1979) and Borovičková and Shimer (2024). Importantly, the innovation is defined relative to the worker's predicted state distribution. It therefore separates the unexpected component of the match realization from systematic differences in expected outcomes across workers. Because the innovation is constructed only from worker-level predetermined characteristics and prior employment history, however, it may also capture a persistent firm-specific component — for instance, if certain firms systematically generate more favorable match realizations for all their workers, independent of worker composition. To isolate the within-firm, time-varying component of match quality from this persistent firm effect, we control for firm fixed effects in the next step, when we aggregate innovations to the firm level and relate them to firm growth, separa-

tion, and survival.

Finally, we aggregate position level innovations within firms and examine their relationship with subsequent employment growth, separation rates, and survival. The firm level regressions include firm fixed effects, so the estimates are identified from changes in realized match quality within the same firm over time. This design allows us to compare a firm after it receives an unexpectedly favorable cohort of matches with the same firm in years when its match realizations are less favorable. We find that favorable match innovations predict faster employment growth, lower separation rates, and a higher probability of survival.

1.1 Literature

This paper contributes to the empirical literature on measuring worker-firm match quality. Because match quality is not directly observed, existing work typically relies on outcomes that should reflect the value of an employment relationship. Belot et al. (2024) provide a recent overview and define match quality broadly as the value generated by a worker-job or worker-employer pairing. They distinguish measures based on worker and employer assessments, direct measures of productivity, wages and tenure, and measures of skill fit, and show that these indicators need not move together. Earlier work often focuses on one or two observable implications of a good match. Bowlus (1995), building on Jovanovic (1979), uses job tenure as an indicator of match quality under the premise that better matches endure, and starting wages as an additional signal of mismatch. She finds that jobs formed in recessions have shorter expected duration and that the effect of labor-market conditions on tenure largely disappears after controlling for starting wages. More recent work uses richer combinations of observables. Mühlbauer and Weber (2024) constructs a predictive measure of match quality from expected job stability and wages using German administrative data and machine-learning methods, while Koenraadt and Timmermans (2026) uses job-to-job flows, subsequent match duration, and promotion rates as outcomes reflecting realized matching quality in Revelio Labs data.

A smaller set of papers attempts to isolate a match-specific component more directly from worker and firm heterogeneity. Jackson (2013) uses student achievement to estimate the productivity associated with teacher-school pairings separately from persistent teacher and school effects. He defines match quality as the fixed, time-invariant productivity associated with a particular worker-firm pairing and shows that a sizable portion of what would otherwise be attributed to teacher quality is school specific. In wage data, Woodcock (2015) extends the worker-firm fixed-effects framework to include a pair-

specific match effect. In his structural interpretation, worker productivity, firm productivity, and match quality enter production separately, and a good match is one whose productivity is high conditional on the worker and firm components. Jinkins and Morin (2018) similarly separates changes in firm wage premia from changes in match quality around job-to-job transitions and finds that improvements in match effects account for a substantial share of movers' wage growth.

Our approach differs from both sets of papers. Rather than selecting a single observed outcome such as tenure or wages as the measure of match quality, or defining match quality as a time-invariant residual in a wage or production equation, we infer a latent match state jointly from multiple outcomes over the worker's employment history. Spell duration and the manner in which a spell ends provide distinct signals about the underlying state of the match. This allows otherwise similar spells to be distinguished according to whether they end in promotion, job-to-job movement, nonemployment, or remain ongoing, and permits the state of the match to evolve across successive employment spells.

We also relate to the broad AKM literature using matched employer-employee data to distinguish persistent worker heterogeneity from persistent firm heterogeneity and to study the assignment of workers across firms (Abowd et al. (1999)). The AKM framework decomposes wages into worker and firm fixed effects and uses worker mobility to identify firm-specific wage premia. Subsequent work uses these components to study wage inequality and assortative matching between high-wage workers and high-wage firms. Match-effect models extend this framework by allowing for a worker-firm-specific component. Woodcock (2015) show that excluding pair-specific heterogeneity can contaminate estimates of both worker and firm effects, and emphasize that a worker who systematically forms high-quality matches with all employers is observationally indistinguishable from a persistently high-quality worker, with an analogous argument for firms. Their normalization therefore reserves the match component for variation specific to a particular worker-firm pairing. Related assortative-matching papers focus instead on whether high-type workers systematically sort toward high-type firms. Shi, for example, identifies sorting between latent worker and firm types using firm profits and worker wages. Our object is the pair-specific component of the employment relationship after allowing workers and firms to differ systematically.

Finally, our empirical approach is related to a growing literature that uses hidden Markov and latent state models to recover persistent unobserved labor market heterogeneity from workers' histories. Shibata (2019) estimates a hidden Markov model in which latent labor market states summarize workers' prior histories and predict subsequent transitions. Ahn et al. (2023) similarly use a hidden Markov model to uncover

segments of the U.S. labor market that differ in employment stability, unemployment exposure, and labor force attachment. More recently, Hong et al. (2026) allow workers' latent wage types to evolve according to a hidden Markov process that depends on the type of firm employing them. Our approach differs from this literature because we estimate latent match states rather than latent worker states. This distinction allows match quality to vary across employment relationships for the same worker and separates the realized quality of a match from persistent worker and firm heterogeneity. Our three step approach then allows us to have a fixed definition of the latent match states and identify the unexpected component of each match realization and examine whether these match specific innovations contribute to subsequent firm growth, employee retention, and survival.

2 Data

Our primary data sources are the Individual Profile Data and Company References from Revelio Labs. Revelio aggregates and standardizes information from LinkedIn professional profiles on a weekly basis, providing detailed information on workers' current and historical employment positions, educational backgrounds, skills, and demographic characteristics. The data allow us to directly observe firm-to-firm transitions, internal moves and promotions within firms, as well as the timing of each position change. Employers are assigned persistent identifiers, which allow employment positions to be linked across workers and over time and enable us to construct longitudinal worker–firm employment histories as well as firm-level workforce panels.

Table 1 compares the distribution of employment across sectors in Revelio Labs with that in the Quarterly Census of Employment and Wages (QCEW). Because LinkedIn users are disproportionately employed in professional and skill-intensive occupations, coverage varies across sectors. Revelio tends to over-represent employment in industries such as information, finance and real estate, professional and business services, and parts of manufacturing, while under-representing sectors such as agriculture, construction, wholesale and retail trade, and health services. Appendix Tables 6–9 report the number and share of unique workers and employers by sector at the annual frequency, benchmarked against the Quarterly Census of Employment and Wages (QCEW). Overall, the coverage of the LinkedIn individual positions data across sectors appears relatively stable over time.

Table 1: Employment distribution by sectors

Source	11 Agr.	21 Min.	22 Util.	23 Constr.	31–33 Manuf.	42 Whole.	44–45 Retail
Revelio	0.2%	0.5%	0.5%	1.6%	9.9%	1.8%	6.0%
QCEW	1.0%	0.5%	0.4%	5.2%	9.7%	4.6%	12.0%
Revelio	175,949	361,782	371,004	1,232,329	7,167,188	1,329,461	4,513,772
QCEW	1,218,570	664,722	570,298	6,715,792	12,380,424	5,811,037	15,223,807
Source	48–49 Trans.	51 Info.	52 Fin.	53 Real Es.	54 Prof. Serv.	55 Mng.	56 Adm. Sup.
Revelio	1.8%	6.8%	7.2%	1.9%	9.0%	0.1%	2.5%
QCEW	4.3%	2.2%	4.6%	1.6%	6.8%	1.7%	6.7%
Revelio	1,346,893	5,115,730	5,289,857	1,495,680	6,724,540	77,523	1,941,206
QCEW	5,565,474	2,780,275	5,844,688	2,101,251	8,737,590	2,188,449	8,506,948
Source	61 Educ.	62 Health	71 Arts	72 Accom.	81 Other Serv.	92 Pub. Adm.	Unclass/Miss. Unclass/Miss.
Revelio	7.7%	6.9%	1.7%	3.0%	3.0%	7.1%	20.7%
QCEW	7.0%	13.3%	1.5%	9.7%	3.4%	3.5%	
Revelio	5,871,309	5,344,179	1,287,927	2,269,546	2,318,170	5,222,737	15,075,350
QCEW	8,919,134	17,045,222	1,964,359	12,428,279	4,343,811	4,453,579	

Notes: This table reports the employment distribution across sectors before any data restriction. *Revelio 1* excludes positions with missing end dates, while *Revelio 2* includes these positions and treats them as ongoing jobs. Employment shares are computed as the fraction of unique active workers within each sector-year and then averaged across years. QCEW reports the numbers from the Quarterly Census of Employment and Wages (<https://www.bls.gov/cew/downloadable-data-files.htm>).

Definition of firms. Revelio provides a higher-level employer identifier, *ultimate_parent_rcid*, which links related employer entities to their ultimate parent company.¹ In our subsequent analysis, we define a firm using *ultimate_parent_rcid*. This definition more closely corresponds to the enterprise-level concept of a firm in the firm-dynamics literature and may combine multiple subsidiaries, operating units, or establishments located in different geographic areas under common ownership. We aggregate employment histories across all constituent *rcids* belonging to the same ultimate parent when constructing firm-level measures of employment, hiring, separations, growth, and survival.

Estimation Sample. Our sample period spans 2007 to 2025, a period during which Revelio provides broad coverage of the U.S. economy. After restricting the data to employment

¹Revelio also provides a more granular identifier, *rcid*, corresponding to an employer entity with a distinct LinkedIn company page. These entities can range from very small, single-person businesses to subsidiaries or operating units of larger corporations, so *rcid* does not map cleanly into either the conventional establishment or firm concept used in the firm-dynamics literature.

spells ending in or after 2007, we first exclude individuals with no education information and those with unusually long education records.² Our estimation sample is then restricted to workers who satisfy two additional criteria. First, we require each worker to have at least three consecutive employment spells recorded in the data, ensuring that we observe a sufficiently informative employment history to characterize the evolution of worker–firm matches. Second, we require workers to be observed for at least one year after age 16, with age inferred from the reported year of birth. After imposing these restrictions, our baseline sample contains 22,874,918 workers, 3,896,164 unique firms, and 113,560,440 worker–firm matches.

Constructing worker–firm employment spells. The unit of analysis is the *worker–firm employment spell*, defined as a contiguous period during which a worker remains employed at the same firm without changing positions. We expand each reported employment spell into monthly observations and retain a single primary position for each worker-month. When a worker holds multiple positions at the same time, we prioritize a position with the same ultimate-parent employer (*ultimate_parent_rcid*) as in the preceding month in order to preserve employment continuity. If no such position exists, we select the overlapping position with the highest reported pay. Finally, months that overlap with an education spell are classified as internships rather than regular employment.

2.1 Measurement of job exit types

A unique feature of this dataset is that it allows us to measure whether workers are moving forward or backward in their careers when they experience job-to-job transitions or internal transfers. All else equal, we interpret a higher-ranked position within a worker’s employment history as indicating a better worker–firm match. These career movements therefore contain valuable information about whether the match-specific component of a worker–firm relationship is improving or deteriorating over time. We first introduce the concept of a seniority ladder, which is central to our classification of career progression. We then describe how we measure the dynamics of workers’ movements along this ladder.

[Briefly mention structural model with career ladders]

Seniority ladder. We primarily rely on Revelio Labs’ seniority variable to infer whether an individual is making advancement or moving backward in their career when experience position change. Revelio Labs constructs an ordinal seniority metric for each

²We define an unusually long education record as a single education spell lasting more than 10 years.

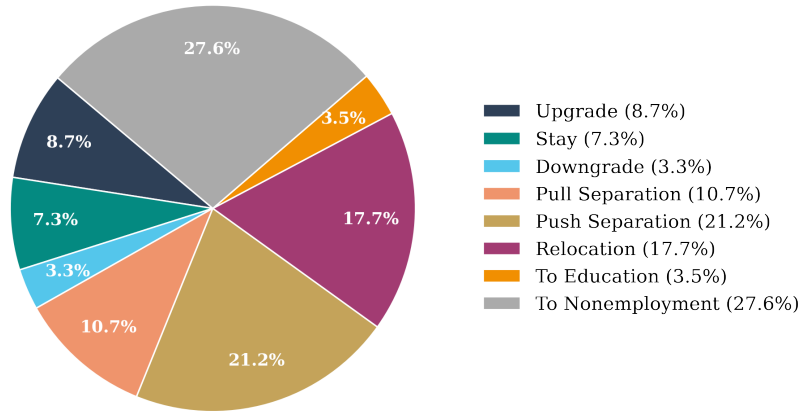
position using three inputs: (1) job title, employer, and industry; (2) the individual's employment history, including prior positions and their associated seniority levels; and (3) the individual's age. These three measures are averaged and benchmarked against the population distribution to produce a metric comprising seven levels, ranging from 1 (entry level) to 7 (senior executive).³ All else equal, higher seniority level during job-to-job transitions could be a signal of higher match quality with the employer.

This seniority ladder, defined over a worker's relative hierarchical position across all positions, is conceptually distinct from the firm-level job ladder studied in the literature. The firm job ladder describes the tendency of workers to move toward more productive and higher-paying firms (Burdett and Mortensen, 1998; Moscarini and Postel-Vinay, 2012, 2016). Empirical evidence strongly supports this pattern: workers systematically move to higher-wage firms during job-to-job transitions, though evidence for a firm-size ladder is more limited (Haltiwanger et al., 2018a,b). Under the firm-ladder definition, all positions within the same employer occupy the same rung. By contrast, workers at the same firm can hold positions at very different seniority levels depending on their roles and responsibilities, generating meaningful within-firm variation in match quality. Although workers are generally expected to climb the seniority ladder over the course of their careers, downward moves during job-to-job transitions are not uncommon: a worker accepting a lower-seniority role at a new employer may nonetheless be moving up the firm-wage ladder if the destination firm is substantially more productive.

Classification into job exit types. We classify the job exit types into one of the nine categories described below. The detailed description about the data process is described in: [8.2](#)

1. *Downgrade*: a move to a lower-seniority position within the same firm.
2. *Stay or internal equal-level transfer*: a move to a position at the same seniority level within the same firm.
3. *Upgrade*: a move to a higher-seniority position within the same firm.

³The seven seniority levels are: (1) Entry Level (e.g., Accounting Intern; Software Engineer Trainee; Paralegal); (2) Junior Level (e.g., Accounts Receivable Bookkeeper; Junior Software QA Engineer; Legal Adviser); (3) Associate Level (e.g., Senior Tax Accountant; Lead Electrical Engineer; Attorney); (4) Manager Level (e.g., Account Manager; Superintendent Engineer; Lead Lawyer); (5) Director Level (e.g., Chief of Accountants; VP of Network Engineering; Head of Legal); (6) Executive Level (e.g., Managing Director, Treasury; Director of Engineering, Backend Systems; Attorney, Partner); (7) Senior Executive Level (e.g., CFO; COO; CEO). For the full methodology, see <https://www.data-dictionary.reveliolabs.com/methodology.html#seniority>.



(a) Distribution of separation types

4. *Pull separation*: a move to a different firm at a higher seniority level.
5. *Push separation*: a move to a different firm at the same or a lower seniority level.
6. *Transition to education*: the worker's next recorded activity is enrollment in an educational program.
7. *Transition to nonemployment*: the worker separates and does not record a subsequent position or educational enrollment within the observation window.
8. *Relocation*: the worker separates from the firm and moves to a different geographic area.⁴
9. *Ongoing*: the position has not ended as of the last data update.

2.2 Descriptive Statistics

PLACEHOLDER Table ?? reports descriptive statistics for the main analysis sample. The sample comprises [X] million workers, [Y] million worker–firm employment spells, and [Z] thousand unique employer firms, with workers observed in an average of [A] spells each. Spell sequences range in length from 1 to [B] spells per worker, with a median of [C] spells.

⁴Geographic areas are defined using the BLS combined MSA classification.

Duration distribution. Spells are distributed across the four duration categories as follows: [X]% are classified as very short (1–6 months), [Y]% as short (7–15 months), [Z]% as medium (16–36 months), and [W]% as long (37 months or more). The concentration of spells in the shorter categories reflects both the reality of high labor market turnover and the tendency of workers to omit short-lived positions from their profiles, potentially understating the true incidence of very short spells.

Exit-type distribution. Among concluded spells, the most common exit type is upgrade ([X]%), followed by pull separation ([Y]%), stay or equal-level transfer ([Z]%), and push separation ([W]%). Downgrades account for [V]% of exits, transitions to education for [U]%, and transitions to nonemployment for [T]%. Ongoing or right-censored spells represent approximately 23% of all spells in the sample.

Pre-spell gap categories. [PLACEHOLDER: frequencies of the pre-spell gap categories, capturing the elapsed time between the end of a worker’s previous spell and the start of the current one.]

Worker characteristics. The sample skews toward college-educated workers: [X]% hold at least a bachelor’s degree, compared with [Y]% in the Current Population Survey for employed individuals over the sample period. Male workers account for [Z]% of the sample; [W]% are predicted non-Hispanic White, [V]% Asian, [U]% Hispanic, and [T]% Black. Occupation categories are dominated by professional and technical roles, consistent with the stronger coverage of Revelio Labs in knowledge-intensive sectors.

Firm characteristics. The median firm in our sample employs [X] workers (as observed in the Revelio Labs data), has been in operation for [Y] years, and is located in one of the top-25 U.S. metropolitan areas. Approximately [Z]% of firms are in professional services or information technology. Employment growth rates average [W]% per quarter, with substantial cross-sectional and time-series dispersion.

Firm growth and survival. Over the sample period, [X]% of firms survive through the end of the observation window. Average employment growth is [Y]% per year, with considerable heterogeneity: [Z]% of firm-years record employment declines. These outcomes are examined in the second-stage analysis in relation to the match-quality distributions recovered from the HMM.

3 A Three-Step Latent Markov Model

Our empirical strategy proceeds in three steps. In the first step, we estimate a multichannel latent Markov model with a small subset of covariates to recover a discrete latent match state for each spell in a worker’s employment history. The latent state is allowed to vary within a worker–firm pair, as changes in positions within the same firm may correspond to changes in the underlying match state. The primary purpose of this step is classification: selecting the number of latent states and characterizing the distinct types of worker–firm matches that they represent. We deliberately limit the covariates included at this stage for two reasons. First, jointly selecting the number of states and estimating a highly parameterized transition model is computationally demanding and may make the resulting classification unstable. Second, this approach gives us a fixed definition of the latent match states based only on employment-spell outcomes and dynamics rather than workers’ demographic characteristics.

In the second step, we recover the posterior probability distribution for each employment spell. We then characterize the recovered states by relating state membership to a comprehensive set of worker and firm characteristics, weighting each spell’s contribution to a given state by its posterior probability from the first-step model. This step allows us to examine which workers and firms are more likely to form each type of match.

In the third step, we construct a spell-level innovation designed to capture the match-specific component of a worker–firm relationship that is not transferable with the worker across firms. To do so, we re-estimate the latent Markov model’s initial-state and transition-probability equations using a richer set of predetermined worker characteristics. These estimates yield, for each spell, a predicted probability distribution over match states conditional on the worker’s observable characteristics and prior employment history. We then compare this predicted distribution with the worker’s realized posterior state distribution to construct an *ex post* match-state innovation. This innovation measures, state by state, the extent to which the realized worker–firm match differs from what would have been predicted based on the worker’s observable characteristics and employment history. Because the predictable component associated with the worker is accounted for, the residual variation is intended to isolate the match-specific component of the realized outcome. In the following section, we aggregate these spell-level innovations to the firm level and examine whether firms that experience unexpectedly favorable match realizations subsequently grow faster and are more likely to survive.

3.1 Step 1: A Multi-Channel Latent Markov Model

In this section, we classify each worker–firm employment spell into one of K discrete latent match states using the joint distribution of spell duration and job-exit type. To do so, we estimate a *multi-channel latent Markov model*.⁵ The model consists of a latent-state process that evolves across a worker’s employment history and an emission process that links the latent state of each spell to its observed outcomes. The observed outcomes can therefore be interpreted as noisy signals of the underlying worker–firm match state. This structure is analogous to a state-space model, in which the latent-state transition process plays the role of the state equation and the emission process plays the role of the measurement equation.

Let $n = 1, 2, \dots, N$ index the worker’s employment spells in chronological order, where the number of observed spells N may vary across workers. Let $U_n \in 1, 2, \dots, K$ denote the latent match state associated with spell n , and let $\mathbf{U} = (U_1, U_2, \dots, U_N)'$ denote the worker’s complete latent-state history. We assume that the latent-state process follows a first-order Markov chain.

We assume that the latent match state follows a first-order Markov process. The economic interpretation of this assumption is that the current latent state summarizes the information of the worker’s entire employment history up until the current spell that are relevant for the state of match formed in the next spell. Thus, conditional on the current match state and observed characteristics of the transition between spells, previous latent states have no additional predictive power for the next state. This assumption allows latent match state to feature persistence and history dependence without requiring the transition process to depend on the worker’s entire employment history. For instance, two workers entering observationally similar jobs may end up with different match states going forward because they arrive from different latent states.

The model is characterized by three sets of probabilities: the emission probabilities, the initial-state probabilities, and the transition probabilities. We describe each in turn.

[Justify why latent match state evolve according to a latent markov chain]

Emission probabilities. Let $\mathbf{Y}_{in} = [Y_{in}^1, Y_{in}^2]'$ denote the vector of observed outcomes for worker i ’s n th match spell. The first emission, Y_{in}^1 , measures *spell duration* and takes four values: 1–6 months, 7–15 months, 16–36 months, and 37 or more months. The second emission, Y_{in}^2 , measures *job-exit type* and takes nine values: downgrade, equal-level internal transfer, upgrade, transition to education, transition to nonemployment, pull separa-

⁵We follow the notation and general framework of the generalized latent Markov model in Bartolucci et al. (2012).

tion, push separation, relocation, and ongoing employment.

For emission $m \in \{1, 2\}$, let $\phi_k^m(y) = P(Y^m = y \mid U_n = k)$, denote the probability of observing outcome y in emission channel m when the underlying match is in latent state k . Here, $U_n \in \{1, \dots, K\}$ denotes the latent match state associated with spell n .

Conditional on the latent-state sequence $\mathbf{U}$, we assume that the response vectors $\mathbf{Y}_1, \dots, \mathbf{Y}_N$ are independent across spells. We further assume that, within a spell, the two emission channels are independent conditional on the current latent state U_n .⁶

[Check the previous paragraph](#)

Initial State Distribution. We parameterize both the initial-state and transition probabilities using multinomial logit specifications. Because different birth cohorts enter the sample at different stages of their careers, we allow the initial distribution of the latent match state to vary across cohorts. Specifically, we model the initial-state probabilities using a multinomial logit specification, with $P(U_1 = k \mid \mathbf{1}) = \frac{\exp(\mathbf{1}'\boldsymbol{\alpha}_k)}{\sum_{k \in K} \exp(\mathbf{1}'\boldsymbol{\alpha}_k)}$, where $\mathbf{1}$ contains indicators for eleven birth-cohorts, including pre-1982, 1982–1984, and subsequent cohorts through 2006–2008, together with an additional category for workers with missing birth-year information.

Transition Probabilities. After the initial spell, the latent match state evolves across successive employment spells. We allow the transition probabilities between latent match states to depend on workers' nonemployment gap preceding each spell. Specifically, we model the transition probabilities using a multinomial logit specification, with $P(U_n = k' \mid U_{n-1} = k, z_n) = \frac{\exp(z_n'\boldsymbol{\beta}_{kk'})}{\sum_{k' \in K} \exp(z_n'\boldsymbol{\beta}_{kk'})}$, where $\mathbf{z}_n$ is a categorical variable indicating the length and nature of the nonemployment gap: no gap, a short gap of one to six months, a long gap of seven or more months spent enrolled in school, or a long gap of seven or more months out of both work and school. Transition probabilities can therefore differ across these four cases, even for workers starting from the same latent state.

Combining the emission model, initial distribution, and transition probabilities, the marginal probability of the observed spell-outcome history $\mathbf{y}$ for a single worker is:

⁶This conditional-independence structure is standard in latent Markov models. It assumes that the latent match state captures the systematic dependence between spell duration and exit type, as well as the persistence in observed outcomes across spells. Conditional on the latent-state sequence, the remaining variation in the observed outcomes is therefore treated as idiosyncratic.

$$P(\mathbf{y}) = \sum_{\mathbf{u}} P(U_1 = u_1 \mid \mathbf{x}) \prod_{t=2}^T P(U_t = u_t \mid U_{t-1} = u_{t-1}, \mathbf{z}_t) \prod_{t=1}^T \phi_{u_t}^1(y_t^1) \phi_{u_t}^2(y_t^2), \quad (1)$$

where the outer sum is over all K^T sequences of latent states. We now reintroduce the worker subscript i to define the sample log-likelihood. For a sample of N workers, the sample log-likelihood is:

$$\ell(\theta) = \sum_{i=1}^N \log P(\mathbf{y}_i), \quad (2)$$

where θ collects all model parameters: the emission probability functions $\{\phi_k^1(\cdot), \phi_k^2(\cdot)\}_{k=1}^K$, the initial-distribution parameters $\{\alpha_k\}_{k=1}^K$, and the transition parameters $\{\beta_{kk'}\}_{k,k'=1}^K$.

Likelihood. Combining the emission model, initial distribution, and transition probabilities, the marginal probability of the observed spell-outcome history $\mathbf{y}$ for a single worker is:

$$P(\mathbf{y}) = \sum_{\mathbf{u}} P(U_1 = u_1 \mid \mathbf{x}_i) \prod_{t=2}^T P(U_t = u_t \mid U_{t-1} = u_{t-1}, \mathbf{z}_{it}) \prod_{t=1}^T \phi_{u_t}^1(y_t^1) \phi_{u_t}^2(y_t^2), \quad (3)$$

where the outer sum is over all K^T sequences of latent states. For a sample of N workers, the sample log-likelihood is:

$$\ell(\theta) = \sum_{i=1}^N \log P(\mathcal{Y}_i), \quad (4)$$

where θ collects all model parameters: the emission probability vectors $\{\phi_k^1, \phi_k^2\}_{k=1}^K$, the initial-distribution parameters $\{\alpha_k\}_{k=1}^K$, and the transition parameters $\{\beta_{kk'}\}_{k,k'=1}^K$. We estimate these parameters by maximum likelihood using the expectation-maximization algorithm. The estimation strategy is described in [Appendix 9.0.1](#).

3.1.1 Number of Latent States and Model Selection

3.2 Step 2: Characterizing the Latent Match States

To understand whether observable worker and firm characteristics systematically relates to the latent match state between the worker-firm pair, we take the fitted HMM parameters from Step 1 as given and combine them with each spell's decoded state information to estimate a spell-level multinomial logit model. This proceeds in two parts: we first construct, for every spell, a full posterior distribution over the K latent match states (Section

??); we then use these posterior probabilities as case weights in a spell-level multinomial logit regression of latent match state on worker and firm characteristics (Section ??).

Soft local decoding Rather than assigning each spell to a single, most-likely match state, we recover the most likely distribution over candidate states.⁷ For worker i 's spell at occasion n , local decoding computes, for every candidate state $u \in \{1, \dots, K\}$, the posterior probability

$$\hat{\pi}_{it}^u \equiv P\left(U_i^{(t)} = u \mid \mathbf{Y}_i\right), \quad (5)$$

conditioning on the worker's entire observed outcome sequence $\mathbf{Y} = (Y^{(1)}, \dots, Y^{(N)})$. This returns a full probability vector $(\hat{\pi}_{it}^1, \dots, \hat{\pi}_{it}^K)$ that sums to one for every spell, rather than a single most-likely state, we refer to this as *soft* local decoding: each spell carries a weight on every candidate latent state, in proportion to how consistent that state is with the worker's observed employment history.

Rewrite

We treat these posterior weights $\hat{\pi}_{it}^u$ as fixed at their Step 1 estimated values throughout the remainder of this section; we do not re-estimate the HMM's transition, emission, or initial-state parameters when constructing the spell-level regressions below.

Spell-level Regression. We relate each spell's soft-decoded posterior distribution over match states to a comprehensive set of predetermined worker and firm observables X_{ijt} , measured as of the start of worker i 's spell t at firm j . X_{ijt} includes bins for worker age, highest educational attainment, ethnicity, firm industry, firm size, firm age, and time fixed effects based on the spell's start quarter.

Let $\pi^k(X_{ijt}; \gamma)$ denote the probability that a spell with characteristics X_{ijt} is of match type k , according to a multinomial logit model:

$$\log \frac{\pi^k(X_{ijt}; \gamma)}{\pi^{\bar{k}}(X_{ijt}; \gamma)} = \gamma_0^k + X_{ijt}' \gamma^k, \quad k \neq \bar{k}, \quad (6)$$

where $\bar{k}$ denotes the reference match state (Stepping-Stone Jobs), normalized so that $\gamma^{\bar{k}} \equiv 0$.

Because we do not observe a spell's true match state, we estimate γ by maximizing the *posterior-weighted* log-likelihood, using each spell's soft local-decoding posterior probabil-

⁷See Bartolucci et al. (2015) about soft local decoding.

ities $\hat{\pi}_{ijt}^k$ from Section ?? as case weights:

$$\hat{\gamma} = \arg \max_{\gamma} \sum_{(i,j,t)} \sum_{k=1}^K \hat{\pi}_{ijt}^k \log \pi^k(X_{ijt}; \gamma). \quad (7)$$

Equivalently, each spell is replicated K times, once per candidate match state, with the k -th replicate assigned outcome k and frequency weight $\hat{\pi}_{ijt}^k$; because $\sum_k \hat{\pi}_{ijt}^k = 1$, each spell contributes a total weight of one observation to the estimation, distributed across states in proportion to its posterior uncertainty rather than concentrated on a single hard-assigned label. This preserves classification uncertainty throughout estimation, rather than first collapsing each spell to its single most likely state and only then relating that label to covariates.

We interpret these estimates as descriptive associations between observable characteristics and match-type membership rather than causal effects of those characteristics on match quality.

3.3 Step 3: Re-Estimation with Covariates and the Construction of Match-State Innovations

In the third step, we hold the soft decoded posteriors fixed, but *re-estimate* the initial-state and transition-probability equations (Sections ?? and ??) using a richer set of predetermined worker characteristics. Because the decoded posteriors from Step 1 are treated as given, this re-estimation does not require re-running the joint EM algorithm on the full model; it instead uses those posteriors as weights in a spell-level weighted likelihood, in the same manner as the multinomial logit of Section 3.2.

3.3.1 An Expanded Covariate Set

Let W_{it} denote an expanded vector of predetermined worker characteristics, measured as of the start of spell t : worker age bins, worker education levels, worker ethnicity, worker gender, labor-market entry cohort, the pre-spell nonemployment-gap category (Section ??).⁸ We re-estimate the initial-distribution equation (??) and the origin-state-specific transition equation (??), replacing $\mathbf{x}$ and $\mathbf{z}_t$ with W_{it} , to obtain updated parameter estimates $\hat{\alpha}_k^W$ and $\hat{\beta}_{kk'}^W$. As in Section ??, the transition equation is estimated separately for

⁸ W_{it} is deliberately richer than the covariate sets used in Sections ?? and ??: those first-step equations use a minimal covariate set so that the classification of match states in Step 1 is not itself sensitive to a large, flexible set of worker characteristics (see Section 9.0.1).

each origin state k .

We estimate $\hat{\alpha}_k^W$ and $\hat{\beta}_{kk'}^W$ by maximizing a posterior-weighted log-likelihood, analogous to equation (7): each worker’s first spell is weighted by its local-decoding posterior $\gamma_1(k)$ in the initial-distribution equation, and each consecutive-spell transition $(t - 1, t)$ is weighted by the product of marginal local-decoding probabilities, $\gamma_{t-1}(k) \gamma_t(k')$, in the origin- k transition equation.⁹ We use local decoding – the full-history posterior – rather than filtered probabilities as the *training* weight because, at the estimation stage, we want the best available estimate of which state a spell actually occupied, and there is no look-ahead concern in fitting a descriptive model of this kind.

3.3.2 Filtered and Smoothed State Probabilities

Constructing a genuine ex ante prediction of a worker’s next match state requires distinguishing between two posterior quantities produced by the forward–backward algorithm (Section 9.0.1). *Local decoding*, $\gamma_t(k) \equiv P(U_t = k \mid \mathbf{y})$, is the smoothed posterior probability of state k in spell t using the worker’s *entire* observed spell history, as defined in Section 9.0.1. *Filtered decoding*, $\alpha_t(k) \equiv P(U_t = k \mid \mathbf{y}_1, \dots, \mathbf{y}_t)$, is instead the posterior probability of state k in spell t using only the worker’s history *up to and including* spell t – it does not use information from any later spell. Both are obtained from the forward pass of the forward–backward recursion; γ_t additionally incorporates the backward pass, which is precisely what allows it to use future spells.

3.3.3 Predicted Match-State Distribution

For each spell t , we construct a predicted distribution over the worker’s match state, $\hat{P}_t(k' \mid \cdot)$, using only information available as of spell $t - 1$ – the worker’s characteristics and employment history up to, but not including, spell t ’s own outcome. For a worker’s first spell ($t = 1$), the predicted distribution is given directly by the re-estimated initial-distribution equation:

$$\hat{P}_1(k' \mid W_{i1}) = P(U_1 = k' \mid W_{i1}; \hat{\alpha}^W), \quad k' = 1, \dots, K. \quad (8)$$

⁹This product weight is a computationally convenient approximation to the true joint smoothed posterior $P(U_{t-1} = k, U_t = k' \mid \mathbf{y})$: it coincides with the joint posterior only under an additional conditional-independence assumption between U_{t-1} and U_t given $\mathbf{y}$, which does not hold exactly in a first-order Markov chain. We view this as an acceptable approximation at the estimation stage, since Step 3’s transition equation is itself an auxiliary, richer-covariate re-fit rather than the primary source of the reported latent-state classification (Section 9.0.1).

For subsequent spells ($t > 1$), the predicted distribution propagates the worker's *filtered* belief about her state at the end of spell $t - 1$, $\alpha_{t-1}(\cdot)$, through the re-estimated transition equation:

$$\hat{P}_t(k' | \cdot) = \sum_{k=1}^K \alpha_{t-1}(k) P(U_t = k' | U_{t-1} = k, W_{it}; \hat{\beta}^W), \quad k' = 1, \dots, K. \quad (9)$$

We use the *filtered* probability α_{t-1} , rather than the smoothed γ_{t-1} , as the origin belief in equation (9) because $\hat{P}_t(\cdot)$ is meant to represent a genuine forecast of spell t 's match state using only information available at the start of spell t . Using γ_{t-1} instead would let information from spells after $t - 1$ – including spell t itself – leak into what is meant to be an ex ante prediction, mechanically shrinking the resulting innovation toward zero.

3.3.4 Match-State Innovation

For each spell t , we define the *realized* match-state distribution as the spell's Step 1 local-decoding posterior, $\gamma_t(k)$ – our best available estimate of the worker's true match state, using the full observed spell history. The *match-state innovation* is the difference between the realized and predicted distributions:

$$\text{Inn}_{ijt}^k = \gamma_t(k) - \hat{P}_t(k | \cdot), \quad k = 1, \dots, K. \quad (10)$$

Because both $\gamma_t(\cdot)$ and $\hat{P}_t(\cdot | \cdot)$ are proper probability distributions over the same K states, $\sum_{k=1}^K \text{Inn}_{ijt}^k = 0$ for every spell: a positive innovation in one state is necessarily offset by negative innovations elsewhere. We interpret Inn_{ijt}^k as the extent to which worker i 's realized trajectory at firm j departed, in spell t , from what her predetermined characteristics and prior employment history would have predicted – an ex post surprise in match state k , purged of information available only after the fact. In the following section, we aggregate Inn_{ijt}^k to the firm level and examine whether firms experiencing unexpectedly favorable match realizations subsequently grow faster and are more likely to survive.

4 Latent Markov Model Estimation Results

4.1 Estimated Latent Match States

In this section, we present results from the estimated multi-channel Markov model. We characterize seven latent match states that capture different stages of the employment trajectory, and describe their emission profiles and transition dynamics, including how

Table 2: Summary of the Seven Latent Match States

State	Labels	Employment-relationship profile	Most likely next states
1	Stepping-Stone Job	Short-duration relationships, with exits concentrated in education (37%) and nonemployment (26%).	Stepping-Stone Job (45%); Churning Job (30%).
2	Stable-Stagnant Match	Long-tenure relationships, but with exits dispersed across nonemployment (21%), push separation (20%), pull separation (12%), and relocation (11%).	Stable-Stagnant Match (37%); Transitional Mismatch (26%); Maturing Growth Match (22%).
3	Churning Job	Short-duration relationships with frequent push separation (26%), nonemployment (22%), and relocation (20%).	Churning Job (35%); Stable-Ongoing Match (28%); Transitional Mismatch (16%).
4	Transitional Mismatch	Medium-duration but separation-dominated relationships, with frequent push separation (33%), nonemployment (24%), and pull separation (19%).	Transitional Mismatch (45%); Stable-Ongoing Match (37%).
5	Early Growth Track	Medium-duration, advancement-oriented relationships; same-level moves and upgrades account for 64% of exits.	Early Growth Track (55%); Stable-Ongoing Match (20%); Transitional Mismatch (10%).
6	Maturing Growth Match	Longer, advancement-oriented relationships; same-level moves and upgrades account for 50% of exits.	Stable-Ongoing Match (54%); Maturing Growth Match (28%); Early Growth Track (12%).
7	Stable-Ongoing Match	Long, ongoing relationships: 99% last at least 37 months and 99.5% remain Ongoing at the end of the observation window.	Stable-Ongoing Match (79%); Early Growth Track (20%).

Notes: The table summarizes the emission and transition profiles of the seven latent states estimated by the multi-channel latent Markov model. The final column reports the most likely destination states for consecutive employment spells with no intervening nonemployment gap. Because State 7 primarily contains ongoing spells, its duration and persistence partly reflect right-censoring.

Note: This figure shows the distribution of all separation types of primary jobs from January 2007 to December 2025. The classification scheme is described in the main text. Source: Revelio Labs

transitions vary with the length and reasons for gaps preceding the start of the next job. We emphasize that underlying match value is a multi-dimensional latent construct, not a simple linear scale: states can differ from one another along duration, exit-type, and mobility dimensions simultaneously, rather than being ranked along a single high-to-low index.

4.1.1 Characterization of the Seven States

Table 3: Estimated Emission Probabilities

Initial distribution	1	2	3	4	5	6	7
	0.582	0.309	0.013	0.036	0.044	0.004	0.012
Emission 1: Employment duration	1	2	3	4	5	6	7
1–6 months	0.429	0.043	0.468	0.048	0.165	0.005	0.009
7–15 months	0.263	0.138	0.371	0.289	0.361	0.050	0.000
16–36 months	0.204	0.298	0.152	0.432	0.423	0.400	0.000
37 months or more	0.103	0.521	0.009	0.231	0.051	0.545	0.991
Emission 2: Job-exit type	1	2	3	4	5	6	7
Education	0.367	0.105	0.094	0.000	0.034	0.017	0.000
Nonemployment	0.257	0.205	0.223	0.240	0.003	0.010	0.000
Downgrade	0.005	0.050	0.010	0.005	0.151	0.108	0.002
Same level	0.044	0.096	0.053	0.010	0.306	0.193	0.002
Upgrade	0.019	0.112	0.048	0.031	0.338	0.306	0.002
Pull separation	0.042	0.120	0.117	0.190	0.047	0.109	0.000
Push separation	0.150	0.203	0.255	0.334	0.087	0.154	0.000
Relocation	0.116	0.109	0.200	0.191	0.034	0.103	0.000
Ongoing	0.000	0.000	0.000	0.000	0.000	0.000	0.995

Notes: This table reports the estimated initial and emission probabilities from the seven-state multi-channel latent Markov model. The reported initial distribution is the weighted average of the cohort-specific initial distributions, where the weights are the cohort shares in the estimation sample. The complete set of cohort-specific initial distributions is reported in the appendix. The first emission is employment duration, and the second is job-exit type. Within each emission panel, a column reports the conditional distribution of the corresponding emission given the latent state. Entries may not sum exactly to one because of rounding. *Sources:* Revelio Labs and authors’ calculations.

Table 2 characterizes the seven states by summarizing their employment-duration, exit-type, and transition profiles. Table 3 reports the initial distribution and emission probabilities for each latent match state, and figure 2 show the transition matrices. The

reported initial distribution is the weighted average of the cohort-specific initial distributions, using each cohort's share in the estimation sample as its weight. The complete set of cohort-specific initial distributions is reported in the appendix.

The emission probabilities reveal substantial heterogeneity across the seven states. Broadly, States 1–4 are relatively separation-prone, whereas States 5–7 are more closely associated with advancement and stable employment. State 1, which we label the *Stepping-Stone Job*, consists primarily of short employment relationships. The initial match-state distributions of later cohorts—whose employment histories are more likely to be observed from labor-market entry—are concentrated in this state. Approximately 43 percent of State 1 relationships last 1–6 months, and another 26 percent last 7–15 months. Exits are concentrated in transitions to education and nonemployment, which account for 37 and 26 percent of exits, respectively. The combination of short duration and frequent movement into education or nonemployment suggests that these jobs often represent early or temporary stages in workers' careers rather than stable relationships that develop within the same firm.

State 2, the *Stable-Stagnant Match*, primarily captures left-censored employment relationships: earlier cohorts often enter the data in the middle of their careers, so their first observed spell may be a mature relationship whose earlier development is unobserved. They are highly stable in duration, with 52 percent lasting at least 37 months, and some subsequently transition into the Maturing Growth Match. At the same time, they remain separation-prone: nonemployment and push separations each account for approximately 20 percent of exits, while pull separations and relocation account for another 12 and 11 percent. State 2 therefore combines long tenure and some potential for subsequent growth with a substantial risk of separation.

State 3, the *Churning Job*, represents short and unstable employment relationships. Nearly 47 percent last only 1–6 months, and another 37 percent last 7–15 months. Push separations, nonemployment, and relocation jointly account for approximately two-thirds of exits, while same-level moves and upgrades are uncommon. These patterns distinguish State 3 from the *Stepping-Stone Job*. Both states contain relatively short relationships, but State 1 is more closely associated with education and early-career transitions, whereas State 3 is more strongly associated with repeated separation and job turnover.

State 4, the *Transitional Mismatch*, resembles the *Churning Job* but is characterized by moderately longer tenure and even greater concentration in separation-related exits. Approximately 43 percent of these relationships last 16–36 months, while 29 percent last 7–15 months. Push separations account for 33 percent of exits, followed by nonemployment at

24 percent, pull separations at 19 percent, and relocation at 19 percent. Despite its high chance of exiting to separation, State 4 differs from the Churning Job in its subsequent dynamics: the transition estimates show that Transitional Mismatches are considerably more likely to be followed by stable employment relationships.

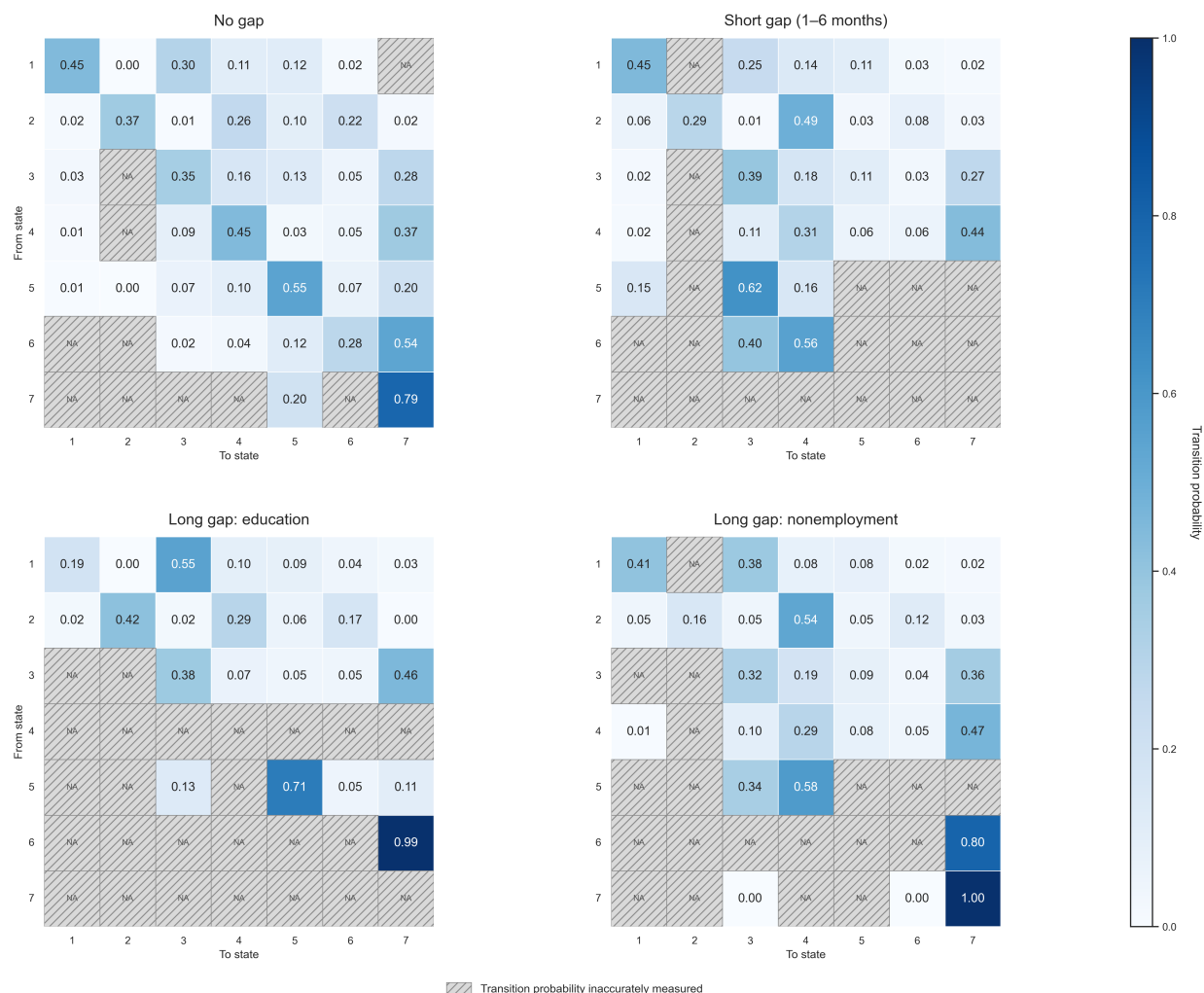
States 5 and 6 describe successive stages of growth-oriented employment relationships. State 5, the *Early Growth Track*, is concentrated at intermediate durations: 36 percent of its matches last 7–15 months and 42 percent last 16–36 months. Same-level moves and upgrades account for 64 percent of exits, the highest combined share among the seven states. Its transition pattern also points toward further career progression: following seamless job transitions, State 5 relationships are likely either to persist or to be followed by States 6 and 7. State 6, the *Maturing Growth Match*, has a distinctly longer tenure profile, with 55 percent of matches lasting at least 37 months and another 40 percent lasting 16–36 months. Moreover, State 6 is much more likely to transition to State 7, the *Stable-Ongoing Match*. Together, these patterns suggest that State 6 lies above State 5 on a career-job ladder. As workers advance into more senior positions, employment relationships take longer to yield further advancement and become increasingly likely to culminate in stable, ongoing matches.

Finally, State 7, the *Stable-Ongoing Match*, consists almost entirely of long and ongoing employment relationships, and captures right-censoring at the end of the sample. Approximately 99 percent last at least 37 months, and 99.5 percent remain ongoing at the end of the observation window. State 7 is consequently close to an absorbing state in the observed data.

4.1.2 Model Robustness Discussion

- Compare AIC and BIC across alternative values of K .
- Conduct out-of-sample validation using predictive likelihood or emission prediction in held-out worker histories.
- Explain the training and validation split.
- Verify that the seven-state solution does not split nearly identical states and that each state has a distinct emission and transition profile.
- Examine stability across samples by re-estimating the model on random worker subsamples.

4.1.3 Transition Dynamics of the Seven States



Notes: Each panel reports the estimated probability of transitioning from the latent match state shown on the vertical axis to the state shown on the horizontal axis, conditional on the type of gap preceding the subsequent employment relationship. Striped cells denote transition probabilities that have large standard errors because the corresponding origin–destination transition have a small number of observations in the estimation sample. Missing cells are especially common for State 7, which consists primarily of ongoing, right-censored employment relationships for which a subsequent state is generally not observed.

Figure 2: Transition Probabilities by Pre-employment Gap

Figure 2 reports the estimated transition matrices separately by the gap between consecutive employment spells. Rows denote the state of the previous spell, while columns denote the state of the subsequent spell. The first panel reports transitions with no intervening gap, the second reports transitions following a short gap of 1–6 months, and the final two panels report transitions following gaps of at least seven months associated with education and nonemployment, respectively. Approximately 65.7 percent of transitions involve no gap, 10.0 percent follow a short gap, 11.5 percent follow a long education-

related gap, and 12.7 percent follow a long nonemployment gap. Shaded cells indicate imprecisely estimated transition probabilities for which the width of the 95 percent confidence interval exceeds 0.20. We calculate standard errors using the delta method, accounting for the nonlinear mapping from the estimated multinomial-logit coefficients to transition probabilities. Appendix 9.1 describes the delta-method calculation and reports the standard errors for the transition matrices.

The no-gap transition matrix displays a clear concentration along the diagonal and in the upper-right portion of the matrix. The large diagonal entries indicate substantial state dependence: workers frequently enter a subsequent employment relationship of the same latent type as their previous relationship. At the same time, transitions above the diagonal are generally more common than transitions below it, indicating that workers tend to move toward higher-numbered states. Since the states are ordered broadly from short, separation-prone relationships to growth-oriented and stable-ongoing relationships, this asymmetry is consistent with an advancing career trajectory. The process is not monotonic, however. We also observe repeated states and movements toward lower-numbered states, suggesting that career progression often involves recurring or transitional steps rather than a one-directional movement up a deterministic ladder.

State 7, the *Stable-Ongoing Match*, is the most prominent destination in the no-gap matrix. It is an especially important destination from State 3 (*Churning Job*), State 4 (*Transitional Mismatch*), and State 6 (*Maturing Growth Match*), with estimated transition probabilities of 0.28, 0.37, and 0.54, respectively. In particular, the high probability of moving from a *Maturing Growth Match* to a *Stable-Ongoing Match* supports the interpretation of State 7 as a frequent endpoint of a growth-oriented career trajectory.

The asymmetry between the upper-right and lower-left portions of the matrix becomes much weaker when there is a gap in employment. Following a short gap, transitions from the two growth-oriented states are concentrated in lower-numbered, separation-prone states. State 5 (*Early Growth Track*) moves primarily to State 3 (*Churning Job*), while State 6 (*Maturing Growth Match*) moves primarily to States 3 and 4 (*Transitional Mismatch*). Thus, an interruption in employment is associated with a break in the otherwise upward career trajectory observed among seamless job transitions. At the same time, transitions from one growth-oriented state directly into another are often imprecisely estimated after a gap. This imprecision reflects the relatively small number of observed cases in which a growth-oriented relationship ends by a nonemployment interval, and then leads immediately to another growth-oriented relationship.

The transition matrix following a long education-related gap is estimated less precisely because this category accounts for a relatively small share of transitions and some

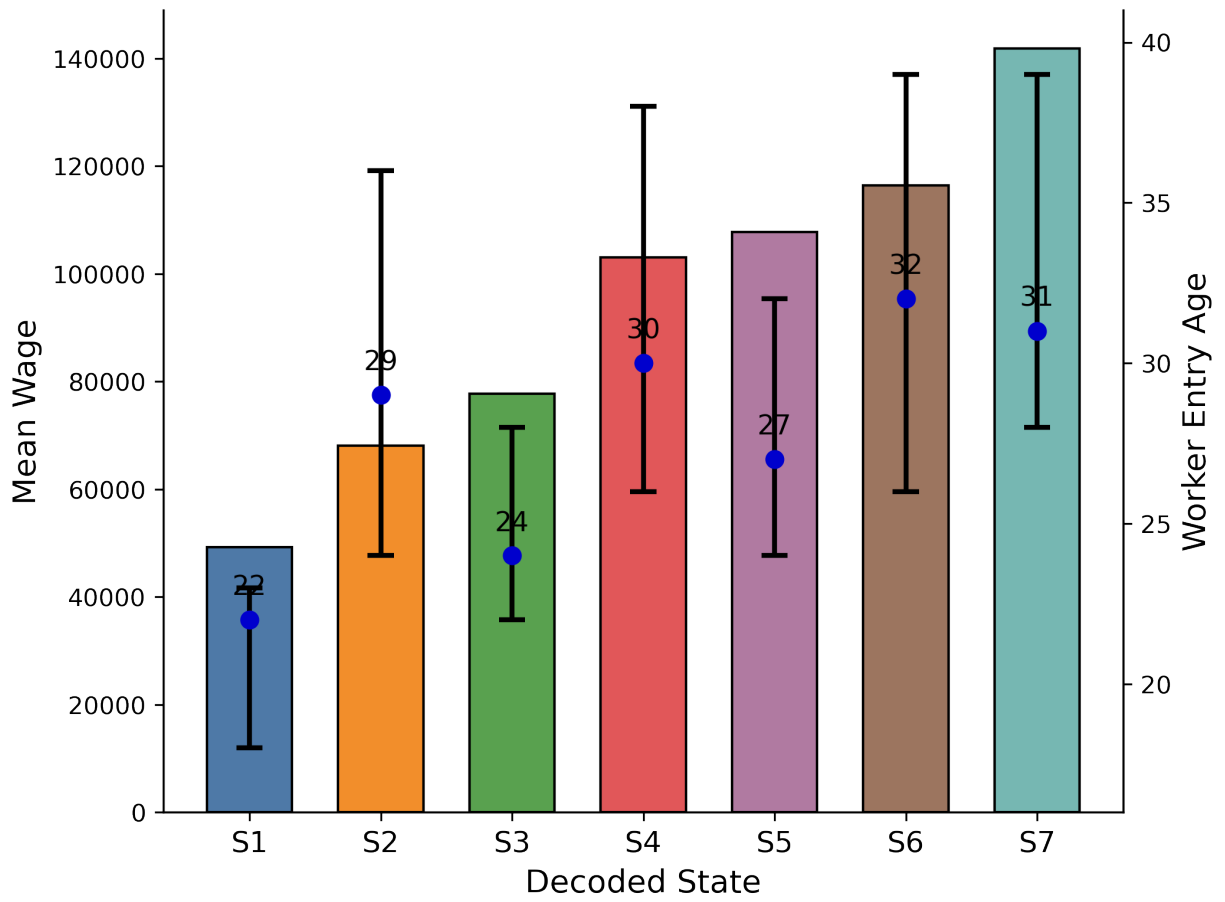
origin-state-gap combinations are rare. In particular, transitions following education are poorly measured for State 4 and State 7. Among the more precisely estimated cells, State 7 remains an important destination. State 6 moves to State 7 with a probability close to one, a pattern consistent with workers using an education spell to move from a maturing growth relationship into a stable-ongoing one. State 5 is highly persistent following education, with a 0.71 probability of returning to the same latent match type, suggesting that education reinforces an early growth trajectory. State 3 (Churning jobs) is another common destination following education, including from State 1 and State 5, which may reflect a period of renewed job search and experimentation as workers attempt to find a suitable post-education match.

Following a long nonemployment gap, the transition distribution is more dispersed and transitions originating from the growth-oriented states are again frequently imprecise. Among the precisely estimated cells, separation-prone states commonly lead to State 4 (Transitional Mismatch), while State 7 (Stable-Ongoing Match) remains the dominant destination from the more mature states. In particular, State 6 (Maturing Growth Match) transitions to State 7 (Stable-Ongoing Match) with an estimated probability of 0.80, and the estimated persistence of State 7 is close to one. These patterns reinforce the interpretation of State 7 as a near-terminal observed state, although its apparent absorbing nature partly reflects right-censoring rather than a structural absence of subsequent mobility.

More generally, imprecision is concentrated among transitions involving origin-state and gap combinations that are rarely observed. It is particularly common for transitions originating from State 7 because this state consists almost entirely of ongoing relationships, leaving only a small and selected subset of spells for which a subsequent state is observed. Transitions involving States 1 and 2 may also be difficult to estimate for some gap categories because these states are disproportionately observed as initial states in workers' recorded histories. Accordingly, the precisely estimated portions of the matrices provide evidence of persistence, career progression, and disruption following employment gaps, while the shaded cells should not be used to draw strong conclusions about rare transition paths.

4.1.4 Descriptive evidence on wages and age by states

Figure 3 provides external validation of the latent-state classification by comparing mean wages and worker ages across states. The bars report mean wages, and the markers and error bars report the median and interquartile range of worker age, respectively. Note that neither wages nor worker age is used to define the latent states, so systematic differences along these dimensions provide an independent check that the estimated states capture



Notes:

Figure 3: Wage and Age distribution by State

economically meaningful differences across matches.

The patterns broadly align with a life-cycle interpretation of the latent states. States associated with workers' initial labor-market experiences tend to have lower mean wages, whereas more mature matches are associated with higher wages. Stepping-Stone Jobs (S2), Churning Jobs (S3), and Early Growth Matches (S5) are disproportionately held by younger workers. This pattern is particularly pronounced for S2, which has a median age of 22 and a comparatively narrow interquartile range. Its strong concentration among very young workers supports the interpretation of S2 as an entry-stage employment state rather than one that is prevalent throughout workers' careers. In contrast, Maturing Growth Matches (S6), Transitional Mismatches (S4), and Stable-Ongoing Positions (S7) have median ages of approximately 30 or above and substantially broader age distributions. In the next section, I examine these relationships more systematically using spell-level multinomial regressions that relate latent-state membership to a broader set of worker and firm characteristics.

The positive relationship between wages and match maturity is also consistent with the interpretation of the states based on their tenure and exit patterns. States characterized by early-career employment, instability, or frequent transitions occupy the lower portion of the wage distribution, while states associated with greater persistence and more advanced career stages tend to pay more. At the same time, the ordering is not perfectly monotonic: workers of similar ages can occupy states with different mean wages, and the age distributions overlap considerably across several states. Thus, the latent states do not merely partition workers by age. Instead, they appear to capture distinct types of employment relationships that are systematically related to workers' career stage and labor-market outcomes.

4.2 Spell-level Multinomial Regression: Who Forms Each Type of Match?

I estimate a spell-level multinomial logit model to characterize how worker and firm attributes correlate with decoded match quality. Let U_{ij}^t denote the decoded match-quality state between worker i and firm j at quarter t . I model the log-odds of being in state k relative to the baseline state, Stepping-Stone Jobs (S1), as a linear function of worker and firm characteristics:

$$\log \frac{P(U_{ij}^t = k \mid X_{ijt})}{P(U_{ij}^t = \text{Stepping-Stone} \mid X_{ijt})} = \gamma_0^k + X_{ijt}' \gamma^k, \quad k \neq \text{Stepping-Stone}, \quad (11)$$

where X_{ijt} includes bins for worker age, worker's highest educational attainment at the start of the match, worker ethnicity, firm industry, firm size, firm age, and time fixed effects. Each element of γ^k gives the change in the log-odds of state k (relative to Stepping-Stone) associated with moving from the reference category to the corresponding covariate bin, holding all other covariates fixed; equivalently, $\exp(\gamma^k)$ is the odds ratio associated with that covariate bin. Tables 11–17 in Appendix 9.2 report the estimated coefficients and odds ratios for each state–characteristic combination.

[Missing: AME]

Because odds ratios are defined relative to a common baseline and to the reference category within each covariate, they support comparisons *within* a covariate—e.g., which age bin is associated with higher odds of a given match-quality state—but they do not directly translate into probability-scale magnitudes, nor are they comparable *across* covariates with different numbers of categories or different baseline prevalences. To quantify magnitudes on the probability scale and to compare the relative importance of different covariates, I complement the coefficient estimates with average marginal effects (AMEs), reported in Table ??.

Figures ?? and ?? plot the estimated odds ratios and 95% confidence intervals from Tables 11–17 as forest plots, grouped by covariate and match-quality state. These figures are designed for within-covariate comparisons—for example, identifying which age or education categories are associated with higher or lower odds of a given match-quality state relative to Stepping-Stone jobs—rather than for comparing effect magnitudes across different covariates.

11 reports the odds ratio for worker age, leaving workers below 20 as the base group. We see that the odds of being in any state relative to the stepping stone job increases with age, consistent with stepping stone jobs focusing on younger workers right out of school. [I should have used a different base group]

Table 12 reports odds ratios for worker gender, with male workers as the reference group. The gender gradient across states is modest in magnitude. Female workers have higher odds of being in the Churning and Early Growth Track states, with odds ratios of 1.029 and 1.057, respectively, and lower odds of being in the Transitional Mismatch and Maturing Growth Track states. The remaining two states show no statistically distinguishable gender differences. Thus, while gender is systematically correlated with decoded match quality, the magnitude of these differences suggests that it is not a first-order determinant of state assignment.

We also find a pronounced pattern for workers with missing gender predictions. Relative to male workers, these workers have lower odds of appearing in every non-baseline

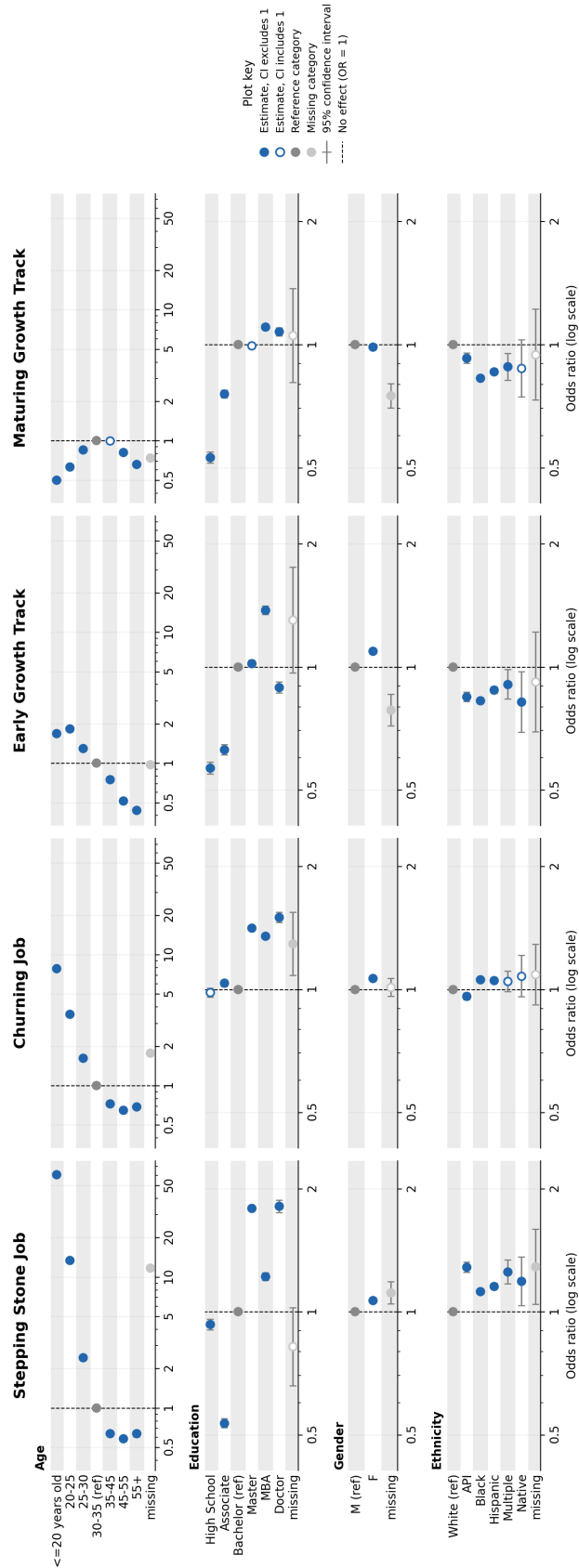


Figure 4: Demographic Characteristics

state, with the reduction ranging from 18 percent in State 6 to 53 percent in State 3.

Table 13 reports odds ratios by workers' highest degree at the time of hire, with bachelor's degree as the reference group. Consistent with our intuition, educational attainment is strongly correlated with the distribution of workers across match-quality states. Relative to bachelor's workers, those with a high school or associate degree are more likely to appear in Churning Jobs, Transitional Mismatches, and Stable-Stagnant Jobs, but substantially less likely to appear in growth-oriented states. The pattern differs for workers with an MBA or master's degree: both groups have relatively higher odds of appearing in the Early Growth Track than in Transitional Mismatches, with this contrast more pronounced among MBA holders.

The distribution is more compressed for workers with doctoral degrees, with one notable distinction from workers with lower degrees: they are the only group for whom Maturing Growth Matches have higher relative odds than both Early Growth Track and Transitional Mismatches. At the same time, doctoral workers still exhibit relatively high odds of appearing in Churning Jobs and Stable-Stagnant Jobs. The relatively high prevalence of Churning Jobs even among the most highly educated workers suggests that job churning and match discovery are not confined to workers with lower educational attainment.

Table 14 reports odds ratios by workers' ethnicity group, with White workers as the reference group. The relative ordering of the latent states is broadly consistent across ethnic groups, suggesting that the overall distribution of match types is similar across groups.

Table 15 reports odds ratios by firm size, with firms with 1–10 employees as the reference group. Firm size is strongly associated with the distribution of match-quality states, with the relationship particularly pronounced for growth-oriented states. As firm size increases, the odds of appearing in the Early Growth Track rise monotonically, from 1.89 for firms with 11–30 employees to 4.75 for firms with more than 200 employees. The odds of appearing in the Maturing Growth Track also increase steadily, from 1.26 to 1.86. In contrast, Churning Jobs exhibit a more moderate positive gradient, while the other match types show much weaker relationships with firm size. Overall, the finding that larger firms are disproportionately represented in growth-oriented states is consistent with the idea that larger firms may provide better career ladders and internal opportunities that support worker and firm growth.

Table 16 reports odds ratios by firm age, with firms younger than one year as the reference group. Similar to the pattern observed for firm size, older firms are more likely to be associated with mature growth-oriented matches and less likely to be associated with

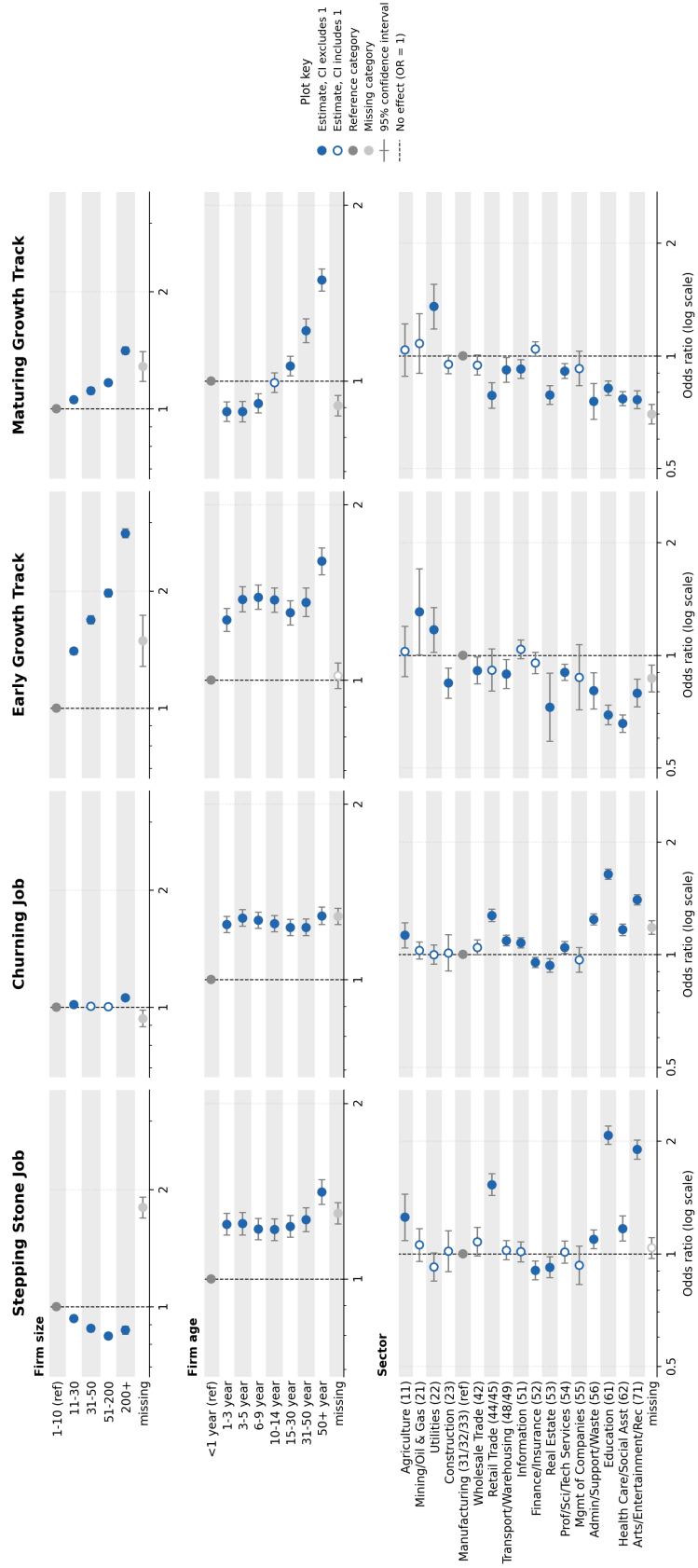


Figure 5: Firm Characteristics

Churning Jobs. The relationship is particularly pronounced for the Maturing Growth Track: its odds increase steadily with firm age, reaching 1.5 among firms aged 50 years or more, even after controlling for firm size. This pattern is consistent with the idea that more established firms may offer more developed internal career ladders and opportunities for advancement, particularly at more senior levels.

Table 17 reports odds ratios by industry, with Agriculture serving as the reference group. Educational Services and Arts, Entertainment, and Recreation stand out for their substantially lower odds in the growth-oriented states. In contrast, Finance and Insurance, Utilities, and Mining, Quarrying, and Oil and Gas Extraction exhibit relatively high odds in growth-oriented states. Utilities has the strongest relative association with these states, with odds ratios of 1.30 in Early Growth Track and 1.55 in Maturing Growth Track.

Professional, Scientific, and Technical Services (54), Information (51), show uniform distribution across latent states. Unlike sectors such as Finance, Utilities, and Mining, these large and diverse sectors do not exhibit strong concentration in particular match-quality states.

4.3 MLMM with full sets of covariates and Calculating innovations

Following the standard three-step estimator of Bartolucci, Montanari, and Pandolfi (2015), we approximate the posterior transition weight for an origin-destination state pair by the product of the marginal posterior probabilities for the adjacent spells. We then estimate origin-specific weighted multinomial logit models. This procedure preserves uncertainty in latent-state classification while avoiding hard assignment of workers to states.

- Summary states of innovations
- Correlation of innovations with firm types

5 Firm Growth and Survival

6 Firm Growth and Survival

Having constructed firm level measures of match state innovations, we now examine whether these innovations predict subsequent firm activity. We first relate lagged match innovations to employment growth, worker separation, and survival while controlling for observable firm characteristics, industry, and year effects. We then introduce firm

fixed effects to account for persistent unobserved differences across firms and identify the relationship using changes in match quality within the same firm over time.

6.1 Controlling for observed firm heterogeneity

Table 4: Firm Outcomes and Lagged Match-State Innovations

	Stock			New Hires		
	Growth	Separation	Survival	Growth	Separation	Survival
inn ₁ (Stepping-st)	−0.037*** (0.001)	0.174*** (0.001)	−0.057*** (0.000)	−0.070*** (0.001)	0.206*** (0.001)	−0.050*** (0.000)
inn ₃ (Churning)	−0.261*** (0.001)	0.634*** (0.001)	−0.207*** (0.001)	−0.248*** (0.001)	0.440*** (0.001)	−0.122*** (0.000)
inn ₄ (Transitional)	−0.123*** (0.000)	0.335*** (0.000)	−0.120*** (0.000)	−0.098*** (0.001)	0.216*** (0.001)	−0.069*** (0.000)
inn ₅ (Early Gr)	0.102*** (0.001)	0.128*** (0.001)	−0.050*** (0.000)	0.065*** (0.001)	0.103*** (0.001)	−0.043*** (0.000)
inn ₆ (Maturing Gr)	0.059*** (0.001)	−0.056*** (0.001)	0.004*** (0.000)	0.104*** (0.001)	−0.123*** (0.001)	0.029*** (0.000)
Reference states	2 and 7	2 and 7	2 and 7	2 and 7	2 and 7	2 and 7
6-digit NAICS	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N (firm-years)	24,179,496	25,073,678	25,188,302	9,784,227	10,058,061	10,083,811

Notes: This table reports estimates of the relationship between match-state innovations and firms' future activity. The first three columns construct match-state innovations using the firm's stock of employees, while the final three columns construct them using the firm's new hires. Within each set of columns, the dependent variables are the DHS symmetric employment growth rate from year $t - 1$ to t , the annualized separation rate, and an indicator for firm survival to year t . States 2 and 7 in the reordered classification are jointly omitted and constitute the reference category. All specifications control for lagged firm size and firm age and include 6-digit NAICS industry and year fixed effects. Standard errors clustered at the firm level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Source: Revelio Labs.*

We estimate the following multinomial logit regression at the firm-level to study the effect of match surprises for firm growth and survival:

$$Y_{jt} = \sum_{s \in \{1,2,3,4,6\}} \delta_s \text{innovations}_{s,j,t-1} + \beta X_{j,t-1} + \zeta_{n(j)} + \eta_t + \varepsilon_{jt}, \quad (12)$$

where Y_{jt} denotes an outcome measuring firm j 's activity in year t . We consider three

outcomes: (1) the firm's annual employment growth rate between $t - 1$ and t ,¹⁰ (2) an indicator for whether the firm survives from $t - 1$ to t ,¹¹ and (3) the firm's annualized worker separation rate in year t .¹²

$\text{innovations}_{s,j,t-1}$ is firm j 's average match state innovation toward state s in year $t - 1$. We construct the measure using either the firm's stock of employees or its new hires during the year. The coefficient δ_s captures how an innovation toward state s , relative to the omitted states, is associated with firm activity in the following year.

We jointly omit states 2 and 7 and estimate coefficients for states 1, 3, 4, 5, and 6. The availability of states 2 and 7 varies systematically over the sample period because of censoring. One state is sparsely represented near the beginning of the panel, while the other is sparsely represented near the end. Including both innovations together with year fixed effects therefore creates substantial collinearity. As a robustness check, we instead omit only one state and estimate coefficients for the remaining six states.

The vector $X_{j,t-1}$ contains lagged firm size and firm age. The specification also includes six digit NAICS industry fixed effects, $\zeta_{n(j)}$, and year fixed effects, η_t . Standard errors are clustered at the firm level. The coefficients are identified from both variation across firms and changes within firms over time.

The interpretation of this specification depends on an important assumption. Suppose firm size, firm age, and detailed industry fixed effects adequately capture persistent differences across firms that are related to both match realizations and future outcomes. Under this assumption, the results in Table 5 indicate that match innovations are strongly related to subsequent firm activity.

Innovations toward separation prone states predict substantially weaker firm outcomes. Churning innovations have the largest adverse coefficients. For the stock of employees, a larger Churning innovation is associated with employment growth that is 0.261 lower, a separation rate that is 0.634 higher, and a survival probability that is 0.207

¹⁰We measure employment growth using the DFH growth rate, defined as

$$g_{jt} = \frac{E_{jt} - E_{j,t-1}}{\frac{1}{2}(E_{jt} + E_{j,t-1})},$$

where E_{jt} denotes firm j 's employment in year t . The growth regression is estimated on firms observed in both $t - 1$ and t .

¹¹For each firm observed in year $t - 1$, the survival indicator equals one if the firm has positive number of employees in year t and zero otherwise. We estimate the survival specification as a linear probability model using all firms observed in $t - 1$.

¹²The separation rate is constructed from worker-level employment spells as the number of worker separations from firm j during year t , annualized relative to the firm's employment. The separation-rate regression is restricted to firms that survive from $t - 1$ to t , consistent with the sample used for the employment-growth regression.

lower. The corresponding coefficients for new hires are also large. Innovations toward Stepping Stone and Transitional matches are associated with slower growth, higher separation rates, and lower survival, although their magnitudes are smaller.

Innovations toward growth oriented states are generally associated with better outcomes. Early Growth innovations predict faster employment growth, although they are also associated with higher separation rates and lower survival. Maturing Growth innovations display a more consistently favorable pattern. For new hires, they are associated with growth that is 0.104 higher, a separation rate that is 0.123 lower, and a survival probability that is 0.029 higher. Among the stock of employees, they also predict faster growth, lower separation, and higher survival.

These estimates suggest that match innovations contain substantial information about firms' subsequent dynamics, conditional on observed firm characteristics. The strongest contrast is between Churning and Maturing Growth matches. Churning innovations predict contraction, worker turnover, and firm exit, while Maturing Growth innovations generally predict expansion, retention, and survival.

6.2 Controlling for firm fixed effects

The previous estimation rests on the strong assumption that observed firm characteristics adequately capture persistent firm heterogeneity. This assumption may fail if some firms have persistently better management, working conditions, technologies, or growth prospects. Such characteristics may improve match outcomes while also affecting future employment growth and survival. The cross sectional coefficients could then partly reflect persistent differences across firms rather than innovations in match quality.

To address this concern, we replace the observed firm controls and industry fixed effects with firm fixed effects. This specification absorbs all time invariant observed and unobserved differences across firms. The coefficients are therefore identified by comparing the same firm in years with different match innovations. Firms observed only once in the relevant estimation sample do not contribute to this variation and are excluded.

The results again a clear ordering across match states. Churning remains the most unfavorable innovation under every specification. For the stock of employees, a larger Churning innovation is associated with growth that is 0.346 lower, a separation rate that is 0.612 higher, and a survival probability that is 0.094 lower. For new hires, the corresponding coefficients are -0.229 , 0.235 , and -0.041 . Thus, even within the same firm, years with unexpectedly more Churning matches are followed by slower growth, greater worker turnover, and a lower probability of survival.

Table 5: Firm Outcomes and Lagged Match-State Innovations

	Stock			New Hires		
	Growth	Separation	Survival	Growth	Separation	Survival
inn ₁ (Stepping-stone)	−0.040*** (0.001)	0.105*** (0.001)	0.003*** (0.000)	−0.047*** (0.001)	0.070*** (0.001)	0.003*** (0.000)
inn ₃ (Churning)	−0.346*** (0.001)	0.612*** (0.001)	−0.094*** (0.001)	−0.229*** (0.001)	0.235*** (0.001)	−0.041*** (0.000)
inn ₄ (Transitional)	−0.146*** (0.001)	0.258*** (0.001)	−0.039*** (0.000)	−0.076*** (0.001)	0.100*** (0.000)	−0.009*** (0.000)
inn ₅ (Early Growth)	0.036*** (0.001)	0.075*** (0.001)	−0.005*** (0.000)	0.010*** (0.001)	0.023*** (0.001)	0.008*** (0.000)
inn ₆ (Maturing Growth)	0.002* (0.001)	0.003*** (0.001)	−0.009*** (0.000)	0.050*** (0.001)	−0.091*** (0.001)	0.010*** (0.000)
Reference states	2 and 7	2 and 7	2 and 7	2 and 7	2 and 7	2 and 7
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N (firm-years)	23,911,695	24,784,527	24,901,908	8,506,274	8,655,303	8,671,221

Notes: This table reports estimates of the relationship between match-state innovations and firms' future activity. The first three columns construct match-state innovations using the firm's stock of employees, while the final three columns construct them using the firm's new hires. Within each set of columns, the dependent variables are the DHS symmetric employment growth rate from year $t - 1$ to t , the annualized separation rate, and an indicator for firm survival to year t . States 2 and 7 in the re-ordered classification are jointly omitted and constitute the reference category. All specifications include firm and year fixed effects. For each outcome, firms with only one observation in the corresponding estimation sample are excluded. Standard errors clustered at the firm level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Source: Revelio Labs.*

Transitional innovations also remain unfavorable. They predict lower growth, higher separation, and lower survival for both the stock of employees and new hires. Stepping Stone innovations are associated with lower growth and higher separation, although their relationship with survival becomes small and positive after firm fixed effects are included.

The results for growth oriented states are more nuanced. Early Growth innovations remain positively associated with employment growth, but they also predict somewhat higher separation rates. Their relationship with survival is small and differs between the stock and new hire measures. For Maturing Growth, the estimates based on new hires remain consistently favorable. A larger innovation toward this state predicts growth that is 0.050 higher, a separation rate that is 0.091 lower, and a survival probability that is 0.010 higher. In contrast, the estimates based on the stock of employees become small after firm fixed effects are included.

Overall, the firm fixed effect estimates confirm that the main results are not driven solely by persistent differences across firms. The clearest and most robust finding concerns Churning matches, which predict worse outcomes even when a firm is compared with itself over time. Favorable Maturing Growth innovations among new hires also predict faster growth, greater retention, and higher survival. These results indicate that changes in the realized quality of a firm's matches contain information about its subsequent performance beyond its persistent characteristics.

7 Conclusion

Match quality is central to theories of job creation, separation, and worker mobility, but it is difficult to observe directly. This paper develops an empirical measure of match quality that combines several outcomes of an employment relationship with information from the worker's prior employment history. Using a matched employer and employee panel from Revelio Labs, we estimate a multichannel latent Markov model that classifies employment spells into seven latent match states based on their duration and exit outcomes. Unlike approaches that recover persistent worker or firm types, our framework allows the quality of a worker's matches to evolve across successive employment relationships.

The recovered states range from short, separation prone positions to stable positions with long durations and opportunities for internal advancement. Although wages are not used to estimate the model, entry wages broadly increase across this ranking, providing external support for the economic interpretation of the states. The states also vary systematically with worker and firm characteristics. Younger and less educated workers are more likely to occupy separation prone states, while larger and older firms are more

likely to form matches associated with stability and internal growth.

We then construct match innovations by comparing each position's realized posterior state distribution with the distribution predicted from the worker's characteristics and employment history before the match. These innovations capture the unexpected component of a match realization. When aggregated to the firm level, they strongly predict subsequent firm activity. Innovations toward Churning matches are followed by slower employment growth, higher separation rates, and lower survival. Innovations toward Maturing Growth matches, particularly among new hires, are associated with faster growth, greater retention, and higher survival. These patterns remain when firm fixed effects are included, indicating that the results are not driven solely by persistent differences across firms.

These findings connect the match specific shocks featured in search models to observed firm dynamics. The composition of a firm's workforce reflects not only whom the firm hires, but also how those employment relationships develop after formation. Unexpected changes in match quality can therefore provide information about future firm expansion, workforce stability, and survival. More broadly, the results suggest that understanding firm dynamics requires attention to the evolution of the employment relationships within the firm, rather than only to worker characteristics, firm characteristics, or the number of matches formed.

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8 Data Appendix

8.1 Distribution of Firms and Employment Across Sectors by Year

8.2 Classification of job exit types

Internal transfers. We first check whether a subsequent position at the same employer begins within 31 days of the end date of the current position. If so, we classify the transition as one of three internal transfer types based on the seniority level of the next position relative to the current one. A *downgrade* captures a move to a lower-seniority position within the same firm, signaling a deterioration in match quality that nonetheless remains above the worker's reservation threshold. A *lateral transfer* denotes a move to a different position at the same seniority level and carries no direct signal about the direction of match quality change. An *upgrade* captures a move to a higher-seniority position and signals an improvement in match quality. We then advance to the next position in the employment spell and repeat this classification until no subsequent qualifying position exists, at which point we treat the final position in the spell as a separation and apply the classification procedure described below.

Job-to-job moves within the same area. A transition is classified as a job-to-job move if a position at a different employer begins within 31 days of the end date of the current position, provided the new position extends beyond that end date. Among same-area job-to-job moves, we distinguish between two types. A *pull separation* is a move to a different employer within the same area at a strictly higher seniority level, indicating that the new match offers higher quality than the previous one. A *push separation* is a move to a different employer within the same area at an equal or lower seniority level. Although a downward move along the seniority ladder does not necessarily mean the previous match quality fell below the reservation threshold, it suggests the new match offers substantially higher utility for the worker to accept a lower rank.

Relocation. We classify a job-to-job move as a relocation if the new position is in a different geographic area and begins within two months of the end date of the current position. Geographic area is defined at the MSA or CSA level.¹³ Internal transfers across regions are not classified as relocation, since they reflect changes in match quality with the same employer rather than a geographic reallocation decision. We interpret relocation as primarily driven by shifts in personal or household preferences rather than a deterio-

¹³We do not classify moves across MSAs within the same CSA as relocation, since CSAs represent regions with overlapping labor and commuting markets. The BLS defines CSAs as groups of metropolitan or micropolitan statistical areas with an employment interchange of at least 15%.

Table 6: Number of Establishments by Year and Sector

Year	Source	11	21	22	23	31-33	42	44-45	48-49	51	52	53	54	55	56	61	62	71	72	81	92	99
		Agric.	Mining	Utilities	Const.	Mfg.	Wholesale	Retail	Transp.	Info.	Finance	Real Est.	Prof. Svcs	Mgmt	Admin	Educ.	Health	Arts	Hosp.	Other	Pub. Admin	Unclass. Missing
2007	Revello	15,030	5,508	7,465	85,388	192,742	74,713	159,934	37,749	99,446	96,365	63,689	285,276	8,275	74,985	96,749	132,930	46,107	69,958	88,752	169,698	3,537
	QCEW	95,203	29,418	15,848	887,529	360,737	626,672	1,047,530	246,527	140,824	489,927	368,875	965,658	48,761	483,194	120,995	713,125	112,900	597,319	1,147,925	71,343	191,725
2008	Revello	17,215	6,128	8,480	93,212	208,135	81,111	179,669	41,363	113,158	105,447	70,594	325,050	9,052	84,514	108,410	147,142	52,339	79,119	99,216	186,249	3,919
	QCEW	94,750	30,702	15,974	878,331	359,103	628,682	1,048,989	247,011	141,685	488,495	364,528	992,124	51,049	462,150	123,594	732,925	113,870	606,756	1,186,086	71,894	198,588
2009	Revello	19,258	6,547	9,542	99,373	221,766	87,032	199,818	44,414	128,291	113,954	76,931	367,964	9,709	94,013	120,068	160,943	58,734	88,327	109,822	203,631	4,356
	QCEW	94,436	31,118	16,173	828,247	350,493	617,849	1,030,743	242,310	141,235	475,406	350,994	998,010	51,571	458,640	124,386	749,426	113,259	609,263	1,220,570	72,398	175,743
2010	Revello	22,079	7,216	10,779	107,729	240,525	94,873	226,008	48,803	145,453	123,751	84,592	416,288	10,777	105,303	133,636	176,923	66,078	100,160	122,308	223,869	4,797
	QCEW	94,484	31,487	16,307	788,985	341,833	613,877	1,022,957	239,563	139,197	464,732	342,002	1,008,602	52,069	462,115	127,924	767,846	112,685	616,964	1,249,369	72,494	171,006
2011	Revello	24,177	7,775	11,514	114,079	254,305	100,488	246,681	52,342	158,519	130,391	90,097	448,913	11,544	113,355	144,418	190,429	71,589	111,138	131,913	239,100	5,251
	QCEW	95,181	32,529	16,329	759,549	336,372	613,845	1,020,233	242,355	138,125	461,203	336,500	1,025,810	53,676	466,221	130,582	783,212	113,116	627,410	1,309,746	72,130	180,628
2012	Revello	26,485	8,353	12,138	121,595	269,866	106,479	269,947	56,740	172,647	137,760	96,382	483,718	12,371	122,042	156,348	204,809	77,809	123,769	142,806	254,962	5,743
	QCEW	96,361	33,965	16,557	743,466	334,039	615,836	1,024,770	245,916	138,532	463,013	337,644	1,049,214	56,146	475,330	133,940	800,069	114,942	639,461	1,285,436	71,539	188,918
2013	Revello	27,832	8,816	12,694	129,166	284,575	112,023	292,092	60,966	186,202	145,234	102,990	518,127	13,307	130,974	167,923	218,705	83,614	136,114	153,141	270,577	6,227
	QCEW	97,810	34,942	16,760	739,923	334,005	619,147	1,031,955	248,826	139,283	463,993	344,225	1,074,459	58,506	484,061	137,104	1,273,293	117,330	647,580	788,031	70,576	183,608
2014	Revello	31,133	9,372	13,467	138,023	301,992	118,438	316,999	66,016	201,759	154,098	120,019	558,386	14,183	141,076	181,662	233,829	89,992	150,111	164,673	289,363	6,756
	QCEW	99,711	35,896	16,892	749,251	337,039	620,881	1,036,609	253,038	144,933	470,209	352,619	1,105,203	60,115	493,419	139,671	1,302,469	120,949	656,934	805,422	70,728	188,060
2015	Revello	30,046	9,794	14,317	147,075	319,945	125,018	344,514	71,194	218,072	164,051	120,019	561,897	15,280	151,855	196,482	250,555	96,875	165,316	177,531	309,187	7,312
	QCEW	101,480	35,826	17,031	758,455	339,872	619,387	1,037,343	258,683	146,282	474,306	360,022	1,138,169	61,688	507,961	144,867	1,341,973	123,942	666,585	817,922	71,077	194,230
2016	Revello	36,546	10,124	15,107	155,558	335,842	130,920	371,022	75,844	233,876	173,380	128,556	644,636	16,152	162,477	211,044	266,209	103,219	179,542	190,439	328,345	7,879
	QCEW	102,973	34,169	17,148	771,310	342,909	615,539	1,039,759	263,038	150,725	478,353	369,245	1,155,618	63,231	521,117	147,728	1,401,074	126,611	678,164	827,358	71,016	230,401
2017	Revello	39,352	10,586	15,926	164,149	351,875	136,782	399,641	81,010	250,279	184,491	138,040	690,373	17,004	173,759	227,053	283,068	109,608	193,228	204,216	348,129	8,351
	QCEW	104,003	32,464	17,321	782,720	345,930	612,334	1,038,669	265,752	156,444	482,572	378,585	1,173,332	64,772	528,655	150,376	1,452,487	129,814	694,573	834,706	70,580	208,895
2018	Revello	42,651	11,116	16,997	173,324	370,490	143,201	430,620	87,320	268,904	196,643	148,053	744,462	18,108	187,253	246,416	301,095	116,543	207,715	219,403	371,118	8,782
	QCEW	105,520	32,464	17,694	806,981	350,659	615,153	1,040,887	274,460	166,607	489,787	396,285	1,217,498	66,941	545,611	154,720	1,508,053	136,543	709,233	850,946	71,169	146,218
2019	Revello	45,664	11,493	18,114	181,235	387,077	148,853	458,602	92,986	286,241	209,039	157,900	796,903	19,152	200,060	265,911	319,543	122,692	220,522	233,723	395,128	9,165
	QCEW	106,500	32,610	17,948	828,425	354,230	615,309	1,040,219	285,012	177,863	495,938	409,866	1,255,858	69,668	562,787	138,392	1,565,040	142,652	722,367	855,660	71,776	155,111
2020	Revello	48,308	11,791	19,227	188,233	401,540	153,439	484,981	98,184	307,778	222,635	167,536	854,408	20,107	212,912	286,048	337,410	127,449	225,083	247,670	417,916	9,397
	QCEW	108,262	31,866	18,408	844,774	357,417	614,286	1,039,293	294,031	192,020	510,396	425,033	1,308,176	73,047	578,811	162,830	1,667,037	146,977	724,289	817,827	72,232	179,495
2021	Revello	50,288	12,066	20,411	194,354	412,945	156,393	504,182	103,501	326,168	235,158	175,990	899,130	20,973	223,262	302,636	352,321	132,429	232,374	258,163	434,997	9,586
	QCEW	109,728	31,093	19,168	869,257	365,513	618,141	1,047,926	312,263	218,801	534,342	443,343	1,413,855	79,750	608,788	169,458	1,716,099	151,608	735,639	835,005	72,295	234,337
2022	Revello	52,145	12,370	21,366	199,484	422,262	158,818	521,146	108,050	343,081	245,021	183,247	942,179	21,696	234,468	318,717	365,355	136,867	239,215	266,771	448,443	9,752
	QCEW	111,707	30,978	20,292	900,420	379,839	634,699	1,055,007	333,813	257,553	559,303	466,119	1,558,055	87,478	650,577	178,702	1,773,095	160,705	757,863	861,187	72,391	345,682
2023	Revello	53,984	12,584	22,398	203,226	430,286	161,187	538,559	111,418	359,895	253,308	189,483	989,574	22,543	246,120	335,085	377,470	140,278	241,083	273,763	457,287	9,778
	QCEW	113,506	30,933	21,415	925,221	391,322	638,779	1,068,227	345,839	279,935	572,931	481,593	1,658,686	94,657	676,793	185,983	1,820,164	168,116	776,697	875,283	72,837	336,977
2024	Revello	55,626	12,772	23,015	205,174	436,655	163,324	554,932	113,183	373,634	259,682	193,604	1,028,960	23,426	254,960	347,378	387,237	142,094	238,522	279,099	460,072	9,703
	QCEW	113,723	30,738	22,077	936,179	396,248	637,175	1,066,813	343,787	288,795	579,030	485,770	1,715,086	100,043	683,561	190,194	1,943,893	173,163	788,069	878,983	72,907	274,636
2025	Revello	56,854	12,909	23,125	204,545	437,701	164,332	549,022	113,559	383,688	262,810	195,218	1,055,559	24,414	260,600	355,713	388,908	141,674	230,506	278,841	460,343	9,508
	QCEW	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
All	Revello	36,814	9,859	15,583	152,891	330,554	127,233	370,967	77,086	239,847	179,654	131,269	665,884	16,214	167,052	221,142	268,152	100,842	170,095	191,697	329,917	7,338
	QCEW	102,519	32,400	17,741	822,168	354,309	620,977	1,040,996	274,568	175,491	497,441	389,625	1,211,856	66,287	534,433	148,969	1,295,071	132,177	680,843	969,303	71,743	210,237

Notes: Each cell reports the percentage of unique workers in a given year employed in each sector. Rows sum to 100 percent. Sources: *Individual positions* files with valid end date from Revello.

Table 7: Number of Workers by Year and Sector

Year	Source	11	21	22	23	31-33	42	44-45	48-49	50	51	52	53	54	55	56	61	Health	62	71	72	81	92	99
		Aggric.	Mining	Const.	Mfg.	Wholesale	Retail	44-45	48-49	Info.	Finance	Real Est.	Prof. Svcs	Mgmt	Admin	25-37	61	Health	62	71	72	81	92	99
2007	Revello	59,269	213,618	185,995	542,960	436,984	603,058	2,021,786	616,087	2,514,070	3,098,836	638,436	3,355,802	30,926	791,443	2,254,378	1,907,212	459,646	876,667	902,890	2,790,705	19,877	8,077,434	
	QCEW	1,165,065	660,278	564,101	7,470,978	13,801,244	5,987,142	15,446,908	5,038,523	3,018,755	5,965,017	2,111,695	7,536,567	1,839,616	8,315,439	8,670,932	13,945,929	17,870,028	11,501,142	4,389,416	4,444,189	216,721	-	
2008	Revello	68,266	244,345	213,229	587,838	4,657,108	738,735	2,269,640	667,299	2,729,526	3,280,733	690,292	3,690,575	35,101	890,216	2,662,457	2,223,477	539,364	1,012,239	1,055,595	3,104,190	23,223	8,988,557	
	QCEW	1,166,399	713,939	574,974	7,035,737	14,349,786	5,954,822	15,244,640	5,000,400	2,973,997	5,829,985	2,072,979	7,720,772	1,895,417	7,923,164	8,979,959	14,354,052	18,111,385	11,349,733	4,434,791	4,515,167	208,318	-	
2009	Revello	76,320	253,715	228,462	601,884	4,684,464	789,935	2,461,639	680,870	2,820,297	3,328,150	726,304	3,840,229	38,168	953,312	3,063,072	2,501,425	607,817	1,113,294	1,178,280	3,392,089	25,959	9,772,801	
	QCEW	1,140,124	641,021	578,042	5,866,072	11,778,517	5,561,625	14,479,860	4,668,250	2,793,101	5,590,122	1,929,878	7,379,469	1,855,139	7,084,602	8,956,657	16,465,784	17,167,776	11,007,713	4,321,928	4,522,592	173,694	-	
2010	Revello	88,967	284,799	252,317	649,073	5,007,680	868,268	2,826,663	740,724	3,104,376	3,577,662	798,188	4,223,680	43,797	1,091,889	3,755,339	2,890,049	709,204	1,294,931	1,358,067	3,744,465	30,982	10,845,514	
	QCEW	1,114,585	651,264	569,343	5,409,062	11,455,600	5,466,213	14,417,382	4,575,633	2,683,163	5,455,488	1,874,874	7,355,400	1,854,778	7,331,721	8,935,733	14,908,595	17,480,010	11,029,398	4,299,245	4,529,882	152,464	-	
2011	Revello	101,428	329,872	275,570	715,074	4,647,677	964,507	3,231,500	822,988	3,401,769	3,859,873	872,853	4,636,575	48,730	1,245,273	4,055,106	3,199,838	801,549	1,002,339	1,527,005	3,984,903	36,282	11,790,746	
	QCEW	1,158,995	729,776	565,878	5,393,207	11,668,417	5,545,555	14,602,592	4,656,840	2,648,137	5,475,771	1,869,031	7,656,658	1,914,436	7,645,388	8,814,066	15,893,532	17,655,456	11,297,213	4,361,236	4,407,927	173,376	-	
2012	Revello	115,080	372,953	296,406	799,763	5,862,604	1,065,399	3,642,960	911,306	3,714,665	4,160,995	962,988	5,058,217	54,402	1,410,432	4,318,111	3,720,844	927,538	1,731,137	1,718,061	4,255,088	41,898	12,816,599	
	QCEW	1,188,505	797,846	565,283	5,306,227	11,778,709	5,656,442	14,801,178	4,741,656	2,643,737	5,533,414	1,902,838	7,795,401	2,003,075	7,920,502	8,791,331	15,486,986	1,840,723	11,685,068	4,513,473	4,375,428	171,660	-	
2013	Revello	1																						

Notes: Each cell reports the percentage of unique workers in a given year employed in each sector. Rows sum to 100 percent. Sources: *Individual positions* files with valid end date from Revelio.

Table 8: Distribution of Establishments by Year and Sector

Year	Source	11	21	22	23	31-33	42	44-45	48-49	51	52	53	54	55	56	61	62	71	72	81	92	99	Missing
		Agric.	Mining	Utilities	Const.	Mfg.	Wholesale	Retail	Transp.	Info.	Finance	Real Est.	Prof. Svcs	Mgmt	Admn	Educ.	Health	Arts	Hosp.	Other	Pub. Admn	Unclss.	Missing
2007	Revelio	0.8%	0.3%	0.4%	4.6%	10.4%	4.0%	8.6%	2.0%	5.4%	5.2%	3.4%	15.4%	0.4%	4.0%	5.2%	7.2%	2.5%	3.8%	4.8%	9.2%	0.2%	2.1%
	QCEW	1.1%	0.3%	0.2%	10.2%	4.1%	7.2%	12.0%	2.8%	1.6%	5.6%	4.2%	11.1%	0.6%	5.2%	1.4%	8.2%	1.3%	6.8%	13.1%	0.8%	2.2%	–
2008	Revelio	0.8%	0.3%	0.4%	4.5%	10.1%	3.9%	8.7%	2.0%	5.5%	5.1%	3.4%	15.8%	0.4%	4.1%	5.3%	7.1%	2.5%	3.8%	4.8%	9.0%	0.2%	2.1%
	QCEW	1.1%	0.3%	0.2%	9.9%	4.1%	7.1%	11.9%	2.8%	1.6%	5.5%	4.1%	11.2%	0.6%	5.2%	1.4%	8.3%	1.3%	6.9%	13.4%	0.8%	2.2%	–
2009	Revelio	0.8%	0.3%	0.4%	4.4%	9.8%	3.8%	8.8%	2.0%	5.6%	5.0%	3.4%	16.2%	0.4%	4.1%	5.3%	7.1%	2.6%	3.9%	4.8%	9.0%	0.2%	2.1%
	QCEW	1.1%	0.4%	0.2%	9.5%	4.0%	7.1%	11.8%	2.8%	1.6%	5.4%	4.0%	11.4%	0.6%	5.2%	1.4%	8.6%	1.3%	7.0%	13.9%	0.8%	2.0%	–
2010	Revelio	0.9%	0.3%	0.4%	4.3%	9.5%	3.8%	8.9%	1.9%	5.8%	4.9%	3.3%	16.5%	0.4%	4.2%	5.3%	7.0%	2.6%	4.0%	4.8%	8.9%	0.2%	2.1%
	QCEW	1.1%	0.4%	0.2%	9.0%	3.9%	7.0%	11.7%	2.7%	1.6%	5.3%	3.9%	11.5%	0.6%	5.3%	1.5%	8.8%	1.3%	7.1%	14.3%	0.8%	2.0%	–
2011	Revelio	0.9%	0.3%	0.4%	4.2%	9.4%	3.7%	9.1%	1.9%	5.8%	4.8%	3.3%	16.5%	0.4%	4.2%	5.3%	7.0%	2.6%	4.1%	4.9%	8.8%	0.2%	2.1%
	QCEW	1.1%	0.4%	0.2%	8.6%	3.8%	7.0%	11.6%	2.7%	1.6%	5.2%	3.8%	11.6%	0.6%	5.3%	1.5%	8.9%	1.3%	7.1%	14.9%	0.8%	2.0%	–
2012	Revelio	0.9%	0.3%	0.4%	4.2%	9.2%	3.6%	9.2%	1.9%	5.9%	4.7%	3.3%	16.5%	0.4%	4.2%	5.3%	7.0%	2.7%	4.2%	4.9%	8.7%	0.2%	2.1%
	QCEW	1.1%	0.4%	0.2%	8.4%	3.8%	6.9%	11.6%	2.8%	1.6%	5.2%	3.8%	11.8%	0.6%	5.4%	1.5%	9.0%	1.3%	7.2%	14.5%	0.8%	2.1%	–
2013	Revelio	0.9%	0.3%	0.4%	4.1%	9.1%	3.6%	9.3%	1.9%	5.9%	4.6%	3.3%	16.6%	0.4%	4.2%	5.4%	7.0%	2.7%	4.3%	4.9%	8.6%	0.2%	2.2%
	QCEW	1.1%	0.4%	0.2%	8.3%	3.8%	7.0%	11.6%	2.8%	1.6%	5.2%	3.9%	12.1%	0.7%	5.4%	1.5%	14.3%	1.3%	7.3%	8.8%	0.8%	2.1%	–
2014	Revelio	0.9%	0.3%	0.4%	4.1%	9.0%	3.5%	9.4%	2.0%	6.0%	4.6%	3.3%	16.6%	0.4%	4.2%	5.4%	6.9%	2.7%	4.5%	4.9%	8.6%	0.2%	2.2%
	QCEW	1.1%	0.4%	0.2%	8.3%	3.7%	6.9%	11.4%	2.8%	1.6%	5.2%	3.9%	12.2%	0.7%	5.4%	1.5%	14.4%	1.3%	7.3%	8.9%	0.8%	2.1%	–
2015	Revelio	0.9%	0.3%	0.4%	4.1%	8.8%	3.5%	9.5%	2.0%	6.0%	4.5%	3.3%	16.6%	0.4%	4.2%	5.4%	6.9%	2.7%	4.6%	4.9%	8.5%	0.2%	2.2%
	QCEW	1.1%	0.4%	0.2%	8.2%	3.7%	6.7%	11.3%	2.8%	1.6%	5.1%	3.9%	12.3%	0.7%	5.5%	1.6%	14.6%	1.3%	7.2%	8.9%	0.8%	2.1%	–
2016	Revelio	0.9%	0.3%	0.4%	4.0%	8.7%	3.4%	9.6%	2.0%	6.1%	4.5%	3.3%	16.7%	0.4%	4.2%	5.5%	6.9%	2.7%	4.6%	4.9%	8.5%	0.2%	2.2%
	QCEW	1.1%	0.4%	0.2%	8.2%	3.6%	6.5%	11.1%	2.8%	1.6%	5.1%	3.9%	12.3%	0.7%	5.5%	1.6%	14.9%	1.3%	7.2%	8.8%	0.8%	2.4%	–
2017	Revelio	1.0%	0.3%	0.4%	4.0%	8.5%	3.3%	9.7%	2.0%	6.1%	4.5%	3.4%	16.8%	0.4%	4.2%	5.5%	6.9%	2.7%	4.7%	5.0%	8.5%	0.2%	2.2%
	QCEW	1.1%	0.3%	0.2%	8.2%	3.6%	6.4%	10.9%	2.8%	1.6%	5.1%	4.0%	12.3%	0.7%	5.6%	1.6%	15.2%	1.4%	7.3%	8.8%	0.7%	2.2%	–
2018	Revelio	1.0%	0.3%	0.4%	3.9%	8.4%	3.2%	9.8%	2.0%	6.1%	4.5%	3.4%	16.9%	0.4%	4.2%	5.6%	6.8%	2.6%	4.7%	5.0%	8.4%	0.2%	2.3%
	QCEW	1.1%	0.3%	0.2%	8.3%	3.6%	6.3%	10.7%	2.8%	1.7%	5.0%	4.1%	12.5%	0.7%	5.6%	1.6%	15.5%	1.4%	7.3%	8.8%	0.7%	1.5%	–
2019	Revelio	1.0%	0.2%	0.4%	3.9%	8.3%	3.2%	9.8%	2.0%	6.1%	4.5%	3.4%	17.0%	0.4%	4.3%	5.7%	6.8%	2.6%	4.7%	5.0%	8.4%	0.2%	2.3%
	QCEW	1.1%	0.3%	0.2%	8.3%	3.6%	6.2%	10.5%	2.9%	1.8%	5.0%	4.1%	12.7%	0.7%	5.7%	1.6%	15.8%	1.4%	7.3%	8.6%	0.7%	1.6%	–
2020	Revelio	1.0%	0.2%	0.4%	3.8%	8.1%	3.1%	9.8%	2.0%	6.2%	4.5%	3.4%	17.2%	0.4%	4.3%	5.8%	6.8%	2.6%	4.5%	5.0%	8.4%	0.2%	2.3%
	QCEW	1.1%	0.3%	0.2%	8.3%	3.5%	6.0%	10.2%	2.9%	1.9%	5.0%	4.2%	12.9%	0.7%	5.7%	1.6%	16.4%	1.4%	7.1%	8.0%	0.7%	1.8%	–
2021	Revelio	1.0%	0.2%	0.4%	3.7%	8.0%	3.0%	9.7%	2.0%	6.3%	4.5%	3.4%	17.3%	0.4%	4.3%	5.8%	6.8%	2.6%	4.5%	5.0%	8.4%	0.2%	2.4%
	QCEW	1.0%	0.3%	0.2%	8.2%	3.5%	5.8%	9.9%	2.9%	2.1%	5.0%	4.2%	13.4%	0.8%	5.8%	1.6%	16.2%	1.4%	6.9%	7.9%	0.7%	2.2%	–
2022	Revelio	1.0%	0.2%	0.4%	3.7%	7.8%	2.9%	9.7%	2.0%	6.4%	4.5%	3.4%	17.5%	0.4%	4.4%	5.9%	6.8%	2.5%	4.4%	5.0%	8.3%	0.2%	2.5%
	QCEW	1.0%	0.3%	0.2%	8.0%	3.4%	5.7%	9.4%	3.0%	2.3%	5.0%	4.2%	13.9%	0.8%	5.8%	1.6%	15.8%	1.4%	6.8%	7.7%	0.6%	3.1%	–
2023	Revelio	1.0%	0.2%	0.4%	3.6%	7.7%	2.9%	9.7%	2.0%	6.5%	4.5%	3.4%	17.7%	0.4%	4.4%	6.0%	6.8%	2.5%	4.3%	4.9%	8.2%	0.2%	2.6%
	QCEW	1.0%	0.3%	0.2%	8.0%	3.4%	5.5%	9.3%	3.0%	2.4%	5.0%	4.2%	14.4%	0.8%	5.9%	1.6%	15.8%	1.5%	6.7%	7.6%	0.6%	2.9%	–
2024	Revelio	1.0%	0.2%	0.4%	3.6%	7.6%	2.9%	9.7%	2.0%	6.5%	4.5%	3.4%	18.0%	0.4%	4.5%	6.1%	6.8%	2.5%	4.2%	4.9%	8.1%	0.2%	2.7%
	QCEW	1.0%	0.3%	0.2%	8.0%	3.4%	5.4%	9.1%	2.9%	2.5%	4.9%	4.1%	14.6%	0.9%	5.8%	1.6%	16.6%	1.5%	6.7%	7.5%	0.6%	2.3%	–
2025	Revelio	1.0%	0.2%	0.4%	3.6%	7.6%	2.9%	9.5%	2.0%	6.7%	4.6%	3.4%	18.4%	0.4%	4.5%	6.2%	6.8%	2.5%	4.0%	4.8%	8.0%	0.2%	2.4%
	QCEW	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
All	Revelio	0.9%	0.3%	0.4%	4.0%	8.7%	3.4%	9.4%	2.0%	6.0%	4.7%	3.4%	16.9%	0.4%	4.2%	5.6%	6.9%	2.6%	4.3%	4.9%	8.6%	0.2%	2.3%
	QCEW	1.1%	0.3%	0.2%	8.6%	3.7%	6.5%	10.9%	2.8%	1.8%	5.2%	4.0%	12.5%	0.7%	5.5%	1.5%	13.2%	1.4%	7.1%	10.2%	0.8%	2.2%	–

Notes: Each cell reports the percentage of unique workers in a given year employed in each sector. Rows sum to 100 percent. Sources: *Individual positions* files with valid end date from Revelio.

Table 9: Distribution of Workers by Year and Sector

Year	Source	11	21	22	23	31-33	42	44-45	48-49	51	52	53	54	55	56	61	62	71	72	81	92	99	Missing
		Agric.	Mining	Utilities	Const.	Mfg.	Wholesale	Retail	Transp.	Info.	Finance	Real Est.	Prof. Svcs	Mgmt	Admin	Educ.	Health	Arts	Hosp.	Other	Pub. Admin	Unclass.	Missing
2007	Revelio	0.2%	0.6%	0.5%	1.5%	12.0%	1.9%	5.6%	1.7%	6.9%	8.5%	1.8%	9.2%	0.1%	2.2%	6.2%	5.2%	1.3%	2.4%	2.5%	7.7%	0.1%	22.2%
	QCEW	0.9%	0.5%	0.5%	6.0%	11.1%	4.8%	12.5%	4.1%	2.4%	4.8%	1.7%	6.1%	1.5%	6.7%	7.2%	11.3%	1.4%	9.1%	3.5%	3.6%	0.2%	-
2008	Revelio	0.2%	0.6%	0.5%	1.5%	11.5%	1.9%	5.6%	1.7%	6.8%	8.1%	1.7%	9.1%	0.1%	2.2%	6.6%	5.5%	1.3%	2.5%	2.6%	7.7%	0.1%	22.3%
	QCEW	0.9%	0.6%	0.5%	5.7%	10.8%	4.8%	12.4%	4.1%	2.4%	4.7%	1.7%	6.3%	1.5%	6.4%	7.3%	11.7%	1.5%	9.2%	3.6%	3.7%	0.2%	-
2009	Revelio	0.2%	0.6%	0.5%	1.4%	10.9%	1.8%	5.7%	1.6%	6.5%	7.7%	1.7%	8.9%	0.1%	2.2%	7.1%	5.8%	1.4%	2.6%	2.7%	7.9%	0.1%	22.7%
	QCEW	1.0%	0.5%	0.5%	5.0%	10.1%	4.8%	12.4%	4.0%	2.4%	4.8%	1.7%	6.3%	1.6%	6.1%	7.7%	12.5%	1.5%	9.4%	3.7%	3.9%	0.1%	-
2010	Revelio	0.2%	0.6%	0.5%	1.4%	10.4%	1.8%	5.9%	1.5%	6.5%	7.5%	1.7%	8.8%	0.1%	2.3%	7.4%	6.0%	1.5%	2.7%	2.8%	7.8%	0.1%	22.6%
	QCEW	1.0%	0.6%	0.5%	4.7%	9.9%	4.7%	12.4%	3.9%	2.3%	4.7%	1.6%	6.3%	1.6%	6.3%	7.7%	12.9%	1.5%	9.5%	3.7%	3.9%	0.1%	-
2011	Revelio	0.2%	0.6%	0.5%	1.4%	10.3%	1.8%	6.1%	1.6%	6.4%	7.3%	1.6%	8.8%	0.1%	2.4%	7.7%	6.2%	1.5%	2.8%	2.9%	7.5%	0.1%	22.3%
	QCEW	1.0%	0.6%	0.5%	4.6%	9.9%	4.7%	12.4%	4.0%	2.3%	4.7%	1.6%	6.4%	1.6%	6.5%	7.5%	12.9%	1.5%	9.6%	3.7%	3.8%	0.1%	-
2012	Revelio	0.2%	0.6%	0.5%	1.4%	10.1%	1.8%	6.3%	1.6%	6.4%	7.2%	1.7%	8.7%	0.1%	2.4%	7.8%	6.4%	1.6%	3.0%	3.0%	7.3%	0.1%	22.0%
	QCEW	1.0%	0.7%	0.5%	4.6%	9.9%	4.7%	12.4%	4.0%	2.2%	4.6%	1.6%	6.5%	1.7%	6.6%	7.3%	12.9%	1.5%	9.8%	3.8%	3.7%	0.1%	-
2013	Revelio	0.2%	0.6%	0.5%	1.4%	9.9%	1.8%	6.4%	1.6%	6.4%	7.1%	1.7%	8.7%	0.1%	2.5%	7.8%	6.6%	1.7%	3.1%	3.0%	7.2%	0.1%	21.7%
	QCEW	1.0%	0.7%	0.5%	4.7%	9.8%	4.7%	12.3%	3.9%	2.2%	4.6%	1.6%	6.6%	1.7%	6.7%	7.2%	13.3%	1.5%	9.9%	3.4%	3.6%	0.1%	-
2014	Revelio	0.2%	0.6%	0.5%	1.5%	9.8%	1.9%	6.4%	1.6%	6.5%	7.0%	1.8%	8.7%	0.1%	2.5%	7.8%	6.7%	1.7%	3.2%	3.0%	7.0%	0.1%	21.3%
	QCEW	1.0%	0.7%	0.5%	4.9%	9.7%	4.7%	12.3%	4.0%	2.2%	4.5%	1.6%	6.6%	1.7%	6.8%	7.1%	13.3%	1.5%	10.0%	3.4%	3.5%	0.2%	-
2015	Revelio	0.2%	0.6%	0.5%	1.5%	9.6%	1.9%	6.5%	1.7%	6.6%	7.0%	1.9%	8.8%	0.1%	2.6%	7.8%	6.9%	1.8%	3.2%	3.1%	6.9%	0.1%	20.8%
	QCEW	1.0%	0.6%	0.4%	5.0%	9.6%	4.6%	12.2%	4.1%	2.1%	4.5%	1.6%	6.7%	1.7%	6.8%	7.0%	13.4%	1.6%	10.1%	3.4%	3.4%	0.2%	-
2016	Revelio	0.2%	0.5%	0.5%	1.6%	9.6%	1.8%	6.4%	1.7%	6.6%	7.0%	1.9%	8.7%	0.1%	2.6%	7.8%	7.1%	1.8%	3.3%	3.1%	6.9%	0.1%	20.5%
	QCEW	1.0%	0.5%	0.4%	5.1%	9.5%	4.5%	12.2%	4.1%	2.1%	4.5%	1.6%	6.7%	1.7%	6.9%	6.9%	13.5%	1.6%	10.2%	3.4%	3.4%	0.2%	-
2017	Revelio	0.2%	0.5%	0.5%	1.6%	9.5%	1.8%	6.4%	1.8%	6.7%	6.9%	2.0%	8.7%	0.1%	2.6%	7.9%	7.2%	1.9%	3.4%	3.1%	6.8%	0.1%	20.2%
	QCEW	1.0%	0.5%	0.4%	5.2%	9.4%	4.5%	12.0%	4.2%	2.1%	4.5%	1.6%	6.8%	1.7%	6.8%	6.8%	13.6%	1.6%	10.3%	3.3%	3.4%	0.2%	-
2018	Revelio	0.2%	0.5%	0.5%	1.7%	9.4%	1.8%	6.3%	1.9%	6.8%	6.9%	2.0%	8.8%	0.1%	2.7%	8.0%	7.4%	1.9%	3.4%	3.2%	6.8%	0.1%	19.7%
	QCEW	0.9%	0.5%	0.4%	5.3%	9.4%	4.4%	11.8%	4.3%	2.1%	4.4%	1.6%	6.9%	1.8%	6.9%	6.8%	13.7%	1.6%	10.3%	3.3%	3.3%	0.1%	-
2019	Revelio	0.3%	0.5%	0.5%	1.7%	9.3%	1.8%	6.2%	1.9%	6.9%	6.9%	2.1%	8.9%	0.1%	2.7%	8.1%	7.5%	1.9%	3.4%	3.2%	6.8%	0.1%	19.3%
	QCEW	0.9%	0.5%	0.4%	5.5%	9.4%	4.3%	11.5%	4.5%	2.1%	4.4%	1.7%	7.0%	1.8%	6.8%	6.7%	13.8%	1.7%	10.3%	3.3%	3.3%	0.1%	-
2020	Revelio	0.3%	0.4%	0.5%	1.8%	9.2%	1.8%	6.3%	2.0%	7.0%	6.9%	2.2%	9.0%	0.1%	2.7%	7.9%	7.6%	1.7%	3.2%	3.2%	6.8%	0.1%	19.4%
	QCEW	1.0%	0.4%	0.4%	5.6%	9.5%	4.4%	11.6%	4.8%	2.1%	4.7%	1.7%	7.4%	1.8%	6.7%	6.8%	14.3%	1.3%	8.7%	3.1%	3.5%	0.1%	-
2021	Revelio	0.3%	0.4%	0.5%	1.8%	9.2%	1.7%	6.2%	2.0%	7.3%	7.0%	2.2%	9.2%	0.1%	2.7%	7.9%	7.8%	1.7%	3.1%	3.2%	6.8%	0.1%	18.9%
	QCEW	0.9%	0.4%	0.4%	5.6%	9.3%	4.3%	11.6%	5.0%	2.1%	4.7%	1.7%	7.5%	1.8%	6.8%	6.6%	14.0%	1.4%	9.2%	3.1%	3.4%	0.1%	-
2022	Revelio	0.3%	0.4%	0.5%	1.8%	9.1%	1.7%	6.0%	2.0%	7.4%	7.0%	2.2%	9.3%	0.1%	2.8%	8.1%	7.8%	1.8%	3.1%	3.2%	6.7%	0.1%	18.6%
	QCEW	0.9%	0.4%	0.4%	5.5%	9.3%	4.3%	11.2%	5.1%	2.2%	4.5%	1.7%	7.6%	1.8%	6.9%	6.5%	13.8%	1.6%	9.7%	3.2%	3.2%	0.2%	-
2023	Revelio	0.3%	0.4%	0.5%	1.8%	9.1%	1.7%	5.7%	2.0%	7.2%	6.9%	2.3%	9.4%	0.1%	2.8%	8.3%	7.9%	1.8%	3.0%	3.3%	6.8%	0.1%	18.7%
	QCEW	0.9%	0.4%	0.4%	5.6%	9.2%	4.4%	11.0%	5.0%	2.1%	4.5%	1.7%	7.6%	1.8%	6.6%	6.5%	14.1%	1.6%	9.9%	3.2%	3.2%	0.2%	-
2024	Revelio	0.3%	0.4%	0.5%	1.9%	9.0%	1.7%	5.5%	1.9%	7.1%	7.0%	2.3%	9.4%	0.1%	2.8%	8.4%	7.9%	1.8%	2.9%	3.3%	6.9%	0.1%	18.8%
	QCEW	0.9%	0.4%	0.4%	5.7%	9.0%	4.3%	10.9%	5.0%	2.0%	4.4%	1.7%	7.5%	1.8%	6.4%	6.6%	14.6%	1.7%	9.9%	3.3%	3.3%	0.2%	-
2025	Revelio	0.3%	0.4%	0.5%	1.9%	9.0%	1.7%	5.3%	1.9%	7.2%	7.1%	2.3%	9.6%	0.1%	2.8%	8.3%	8.0%	1.8%	2.7%	3.3%	6.9%	0.1%	18.9%
	QCEW	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
All	Revelio	0.2%	0.5%	0.5%	1.6%	9.9%	1.8%	6.0%	1.8%	6.8%	7.2%	1.9%	9.0%	0.1%	2.5%	7.7%	6.9%	1.7%	3.0%	3.0%	7.1%	0.1%	20.6%
	QCEW	1.0%	0.5%	0.4%	5.2%	9.7%	4.6%	12.0%	4.3%	2.2%	4.6%	1.6%	6.8%	1.7%	6.7%	7.0%	13.3%	1.5%	9.7%	3.4%	3.5%	0.2%	-

Notes: Each cell reports the percentage of unique workers in a given year employed in each sector. Rows sum to 100 percent. Sources: *Individual positions* files with valid end date from Revelio.

ration in the prior employment match.

Transition into education. A position is classified as transitioning into education if an education record begins within three months of the position’s end date. Because enrollment decisions are typically planned in advance, this transition does not necessarily reflect declining match quality with the previous employer and is therefore treated as a separate category.

Transition into nonemployment. A position is classified as transitioning into nonemployment if none of the above conditions are met: there is no same-area position beginning within one month, no different-area position beginning within two months, and no education record beginning within three months. A transition into nonemployment signals that match quality with the previous employer has fallen below the quality threshold.

Figure 6a shows the distribution of transition types for all completed primary positions. Overall, 27.6% of separations end in nonemployment. This share is higher than expected, partly because users tend to omit jobs that could signal poor performance or low ability. We do not view this as a major concern for our estimates, since these missing transitions likely also reflect very low match quality with the previous employer, similar to true transitions into nonemployment.

We observe push separations (21.2%) at nearly twice the rate of pull separations (10.7%). This pattern may seem counterintuitive. Why would more job-to-job transitions ended in equal or lower seniority roles? There are two potential explanations for this: First, workers may rationally accept a lower-seniority position at a more productive firm, tolerating short-term wage losses in exchange for better long-run career prospects.¹⁴ Second, push separations likely over represent secondary-sector workers, who experience high churn and rarely climb the job ladder in a linear fashion. For these workers, match quality is highly dispersed across employers, so most matches are short-lived, and early-career workers in particular tend to move laterally across jobs while searching for a good fit (Ahn et al. (2023), Gregory et al. (2025)). In both cases, a push separation either indicates the previous match have substantially fallen in quality, or the new match have much higher match quality.

¹⁴A recent case illustrates this dynamic well. Andrej Karpathy — former AI director at Tesla and founding research scientist at OpenAI — joined Anthropic as a senior research leader after founding the AI education startup Eureka Labs. His transition from CEO of Eureka Labs to a research role at Anthropic would be classified as a push separation (though this specific move falls outside my sample, which ends in December 2025; see <https://www.reuters.com/business/autos-transportation/former-tesla-ai-executive-openai-founding-member-andrej-karpathy-joins-anthropic-2026-05-19/>). Karpathy’s decision suggests that his match quality at Anthropic is likely higher than it was at Eureka Labs at the time of the change, making the transition productively rational despite the nominal demotion in title.

Table 10: Salary Change and Tenure by Separation Type

	(1) pct	(2) pct	(3) pct	(4) pct	(5) USD	(6) pct
<i>Main effects</i>						
Upgrade	0.345*** (0.003)	0.346*** (0.003)	0.352*** (0.003)	0.350*** (0.003)	21821.9*** (110.1)	0.314*** (0.006)
Downgrade	−0.238*** (0.003)	−0.239*** (0.003)	−0.232*** (0.002)	−0.232*** (0.003)	−20299.8*** (93.2)	−0.207*** (0.004)
Pull sep.	0.448*** (0.004)	0.443*** (0.004)	0.417*** (0.003)	0.416*** (0.004)	19501.9*** (82.3)	0.367*** (0.007)
Push sep.	−0.085*** (0.003)	−0.089*** (0.003)	−0.096*** (0.003)	−0.098*** (0.003)	−9857.1*** (106.9)	−0.098*** (0.004)
Relocation	0.120*** (0.004)	0.116*** (0.004)	0.102*** (0.005)	0.102*** (0.005)	473.7*** (101.8)	0.078*** (0.012)
<i>× USREC</i>						
Upgrade × Rec.				0.030*** (0.010)	−458.9 (350.0)	
Downgrade × Rec.				−0.006 (0.005)	601.8* (310.5)	
Pull sep. × Rec.				0.017** (0.008)	58.7 (315.4)	
Push sep. × Rec.				0.031 (0.026)	−184.8 (352.0)	
Relocation × Rec.				0.006 (0.009)	720.0 (459.0)	
<i>× UNRATE</i>						
Upgrade × UR						0.007*** (0.001)
Downgrade × UR						−0.004*** (0.001)
Pull sep. × UR						0.009*** (0.001)
Push sep. × UR						0.000 (0.001)
Relocation × UR						0.004** (0.002)
Year-month FE	No	Yes	Yes	Yes	Yes	Yes
Sector FE	No	No	Yes	Yes	Yes	Yes
Observations	80,005,113	80,005,113	71,065,892	71,065,892	71,065,892	71,060,757
R ²	0.001	0.001	0.001	0.001	0.109	0.001

Notes: The dependent variable in columns (1)–(3) is the salary change in USD, defined as the start salary of the next position minus the end salary of the current position. Column (4) uses a random sample. The baseline group is *Stay* (same employer, same seniority level). All specifications include a constant (not reported). Standard errors (in parentheses) are two-way clustered by worker and year-month. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

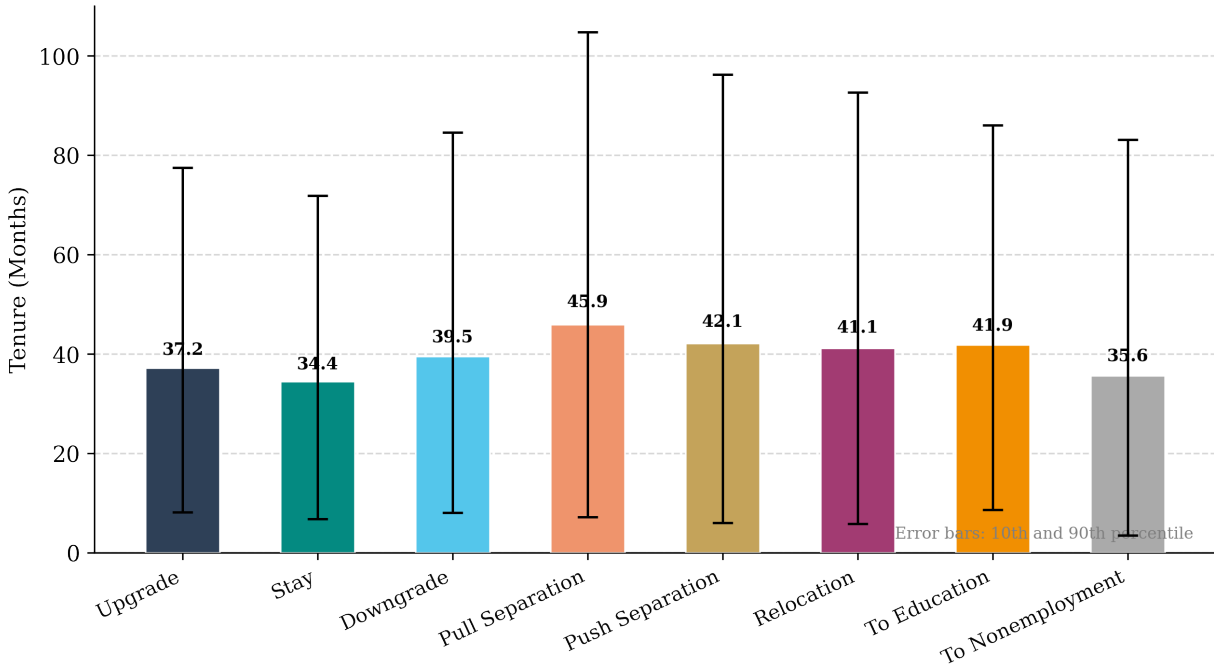
Internal transfers across the seniority ladder are associated with larger salary changes than job-to-job moves across firms. As shown in Figure 6a, 19.3% of position changes are classified as internal transfers, with promotions (8.7%) and lateral transfers (7.3%) occurring more frequently than downgrades (3.3%). This distribution is consistent with the finding of Forsythe (2023) using CPS and SIPP data, who find that over 40% of moves are to lower ranked occupations. Although downgrades are relatively rare, they are associated with substantial salary losses, averaging \$17,400 in annual salary — more than double the average decline of \$8,125 observed for push separations. Promotions, by contrast, are associated with an average annual salary gain of \$26,720, meaningfully larger than the \$23,349 average gain for pull separations ?? Table 10 shows these patterns are robust to controlling for year-month (Column 2) and sector fixed effects (Column 3), indicating that the pattern is not driven by aggregate macroeconomic conditions or by the concentration of different transition types in particular industries.

Turning to cyclical patterns, salary changes across all transition types decline during recessions, but internal promotions and push separations are most affected. Relative to normal times, salary growth associated with promotions falls by \$1,789 during recessions, and push separations experience an additional decline of \$1,766. Notably, during recessions the salary gains from promotions and pull separations become statistically indistinguishable, suggesting that the advantage of internal advancement over cross-firm mobility compresses during downturns.

8.3 Worker and Firm Covariates

The Revelio Labs data provide a rich set of observable worker and firm characteristics, which we use in the descriptive second-stage analysis to examine the determinants of the latent match-quality states recovered by the LMM. These variables are not included in the baseline LMM, which is estimated on the two emission channels alone to preserve the model’s interpretive parsimony.

Worker characteristics. For each worker we observe highest level of educational attainment (high school or below, some college, bachelor’s degree, master’s degree, professional or doctoral degree), which is drawn from the Revelio Labs education file. Age at the time of hire is inferred from the year of the worker’s earliest recorded educational enrollment. Gender is predicted from the worker’s name using a classification algorithm, and predicted ethnicity (White, Black, Hispanic, Asian, or other) is similarly inferred from name and, where available, location signals. We also observe each worker’s



(a) Job duration by separation types

Note: This figure shows the mean, 10th percentile, and 90th percentile of job duration for primary jobs, grouped by the type of subsequent separation. Source: Revelio Labs

occupation category (mapped to the Revelio Labs jobs taxonomy), the geographic location of the position (state and metropolitan area), and the worker’s prior employment history—including the number and types of prior spells, prior seniority progression, and prior separations—which we summarize in a set of pre-spell history variables. All worker characteristics are measured as of the start of each spell.

Firm characteristics. For each employer we observe total employment (headcount), firm age (years since the earliest position recorded in the data), industry sector, and primary metropolitan area of operations. Where financial data are available from Compustat (for publicly listed firms), we supplement these with balance-sheet and income-statement characteristics. We also observe time-varying organizational characteristics such as the firm’s recent employment growth rate and its recent hiring and separation rates, which serve as proxies for firm-level labor demand conditions.

8.4 Firm-Level Outcomes

To examine how match quality relates to firm performance, we construct a panel of firm-quarter observations tracking firm-level employment and survival outcomes. *Firm em-*

employment in a given quarter is defined as the number of workers with an active spell at the firm during that quarter. *Employment growth* is the quarter-on-quarter log change in employment. *Firm survival* is an indicator equal to one if the firm has at least one active position recorded in the data in each quarter through the end of the observation window.

Because Revelio Labs data are derived from online profiles, coverage of firms is inherently incomplete and unequal. Large firms with a strong online presence are well represented, whereas small or private firms with few workers on professional networking platforms may have only a handful of observed positions—or none at all. Observed employment therefore understates true employment, and the degree of understatement varies across firms and over time. We account for this by including firm fixed effects in all firm-level regressions and by focusing on within-firm variation in employment growth and survival rather than cross-sectional levels.

Where relevant, we supplement the employment panel with information from Revelio Labs job postings. *Expected hires* is a forward-looking measure constructed from the stock of active job postings, scaled by historical offer-acceptance rates, and serves as a proxy for the firm’s intended employment growth. This variable allows us to distinguish between realized employment changes and planned hiring activity.

9 Multi-channel Latent Markov Model Appendix

9.0.1 Estimation

We maximize the log-likelihood (4) using the *EM algorithm*. Direct maximization of (4) is computationally demanding because equation (3) sums over K^T latent-state histories. The EM algorithm alleviates this difficulty by alternating between two steps:

- *E-step*: Compute the posterior distribution of the latent-state history given the current parameter estimates, leveraging the Markovian structure through the Baum–Welch (forward–backward) algorithm. This reduces the computational complexity from $O(K^T)$ to $O(K^2T)$.
- *M-step*: Update all parameters by maximizing the expected complete-data log-likelihood given the posterior state probabilities from the E-step. (See Appendix for more details.)

The algorithm iterates until the relative change in log-likelihood falls below 10^{-5} . Because the EM algorithm can converge to local optima, we estimate each model from five independent random restarts and retain the solution with the highest log-likelihood. Each

restart is run as an independent task on a high-performance computing cluster and assigned a distinct seed to ensure reproducibility.

After estimation, we recover two complementary characterizations of each worker’s latent-state history from the fitted model. The first is *global decoding*: the single most likely state sequence for the worker, obtained via the Viterbi algorithm (??).¹⁵ The second, which we refer to as *local decoding*, is the smoothed marginal posterior probability that a worker occupies state k in spell t , $P(U_t = k \mid \mathbf{y})$, obtained directly from the forward–backward recursion used in the E-step, and computed using the worker’s *entire* observed spell history rather than a single jointly-optimal path.¹⁶

Because local decoding preserves the full posterior distribution over match types for every spell – rather than collapsing each spell to a single hard state assignment – we use local decoding, rather than global decoding, as the spell-level weight in the second-step analysis described in Section 3.2: each spell’s contribution to characterizing a given state is weighted by that spell’s local-decoding posterior probability of membership in the state.

The model is estimated on a training sample of one million workers drawn from the Revelio Labs data. The remaining workers are used for out-of-sample validation, as described in Section ??.

9.1 Standard errors for transition probabilities.

We compute standard errors for the estimated transition probabilities using the delta method. The latent Markov model parameterizes each row of the transition matrix as a multinomial logit, so the transition probabilities are nonlinear functions of the estimated logit coefficients. Let $\hat{\gamma}$ denote the vector of transition coefficients and let $\hat{\mathbf{V}}_{\gamma}$ denote its estimated covariance matrix, obtained from the inverse observed information matrix. For a given category, the corresponding transition probabilities can be written as

$$\hat{\pi} = g(\hat{\gamma}; \mathbf{x}),$$

¹⁵In the hidden Markov model literature, global decoding is the procedure most commonly referred to simply as “decoding.” Additional details are in the appendix.

¹⁶Unlike global decoding, local decoding does not require that the state assignments across a worker’s spells form a single valid sequence; it instead reports, spell by spell, the model’s full posterior distribution over match type.

where $\mathbf{x}$ contains the covariates of the model. We approximate the covariance matrix of these probabilities using

$$\hat{\mathbf{V}}_{\pi} = \mathbf{G} \hat{\mathbf{V}}_{\gamma} \mathbf{G}', \quad \mathbf{G} = \left. \frac{\partial g(\gamma; \mathbf{x})}{\partial \gamma'} \right|_{\gamma = \hat{\gamma}}.$$

The reported standard error of each transition probability is the square root of the corresponding diagonal element of $\hat{\mathbf{V}}_{\pi}$. By explicitly incorporating the Jacobian of the multinomial logit transformation, this procedure properly propagates the covariance among the estimated coefficients into the covariance of the transition probabilities, while respecting the adding-up constraint that each row of the transition matrix must sum to one.

9.2 Spell-level regression coefficients

Table 11: Worker age odds ratios by latent state

	S2	S6	S4	S3
≤ 20	60.277 (0.843)	7.782 (0.064)	1.680 (0.031)	0.501 (0.007)
20–25	13.398 (0.138)	3.494 (0.020)	1.819 (0.019)	0.630 (0.004)
25–30	2.424 (0.020)	1.612 (0.009)	1.288 (0.008)	0.852 (0.004)
35–45	0.638 (0.007)	0.726 (0.005)	0.750 (0.006)	0.993 (0.006)
45–55	0.581 (0.009)	0.645 (0.006)	0.515 (0.007)	0.815 (0.007)
55+	0.636 (0.018)	0.686 (0.012)	0.436 (0.009)	0.657 (0.010)
Missing	11.713 (0.198)	1.808 (0.024)	0.798 (0.015)	0.604 (0.010)

Table 12: Gender odds ratios by latent state

	S2	S6	S4	S3
F	1.065 (0.007)	1.064 (0.005)	1.091 (0.007)	0.987 (0.005)
Missing	1.112 (0.036)	1.068 (0.024)	1.188 (0.035)	1.017 (0.025)

Table 13: Education odds ratios by latent state

	S2	S6	S4	S3
High School	0.931 (0.014)	0.982 (0.013)	0.565 (0.010)	0.529 (0.009)
Associate	0.534 (0.007)	1.035 (0.010)	0.627 (0.009)	0.756 (0.008)
MBA	1.218 (0.014)	1.348 (0.010)	1.376 (0.017)	1.103 (0.010)
Master	1.789 (0.013)	1.412 (0.009)	1.019 (0.010)	0.992 (0.008)
Doctor	1.810 (0.032)	1.500 (0.022)	0.891 (0.014)	1.074 (0.012)
Missing	0.821 (0.092)	1.291 (0.117)	1.301 (0.198)	1.052 (0.142)

Table 14: Ethnicity odds ratios by latent state

	S2	S6	S4	S3
API	1.286 (0.019)	0.961 (0.009)	0.844 (0.011)	0.926 (0.014)
Black	1.120 (0.008)	1.055 (0.006)	0.827 (0.007)	0.828 (0.006)
Hispanic	1.154 (0.009)	1.050 (0.007)	0.879 (0.009)	0.858 (0.007)
Multiple	1.252 (0.043)	1.045 (0.031)	0.907 (0.039)	0.882 (0.034)
Native	1.187 (0.083)	1.078 (0.064)	0.821 (0.072)	0.874 (0.072)
Missing	1.287 (0.139)	1.087 (0.095)	0.919 (0.132)	0.944 (0.123)

Table 15: Firm size odds ratios by latent state

	S2	S6	S4	S3
11–30	0.931 (0.007)	1.015 (0.006)	1.399 (0.014)	1.053 (0.009)
31–50	0.878 (0.008)	1.006 (0.007)	1.685 (0.020)	1.112 (0.011)
51–200	0.840 (0.007)	1.002 (0.006)	1.973 (0.022)	1.167 (0.011)
200+	0.870 (0.010)	1.057 (0.008)	2.811 (0.037)	1.410 (0.015)
Missing	1.797 (0.057)	1.210 (0.027)	1.458 (0.053)	1.271 (0.038)

Table 16: Firm age odds ratios by latent state

	S2	S6	S4	S3
1–3 year	1.240 (0.027)	1.242 (0.020)	1.268 (0.029)	0.885 (0.017)
3–5 year	1.243 (0.028)	1.275 (0.022)	1.376 (0.035)	0.885 (0.018)
6–9 year	1.217 (0.026)	1.264 (0.020)	1.388 (0.034)	0.915 (0.018)
10–14 year	1.214 (0.026)	1.247 (0.020)	1.371 (0.034)	0.992 (0.020)
15–30 year	1.229 (0.027)	1.228 (0.020)	1.304 (0.031)	1.060 (0.021)
31–50 year	1.263 (0.030)	1.228 (0.021)	1.360 (0.039)	1.218 (0.029)
50+ year	1.409 (0.035)	1.285 (0.023)	1.598 (0.044)	1.490 (0.033)
Missing	1.296 (0.028)	1.241 (0.020)	1.299 (0.030)	1.098 (0.021)

9.3 Firm Activity Regression on Match Innovations

Table 17: Sector odds ratios by latent state

	S2	S6	S4	S3
Admin and Waste Management and Remed Serv [NAICS 56]	1.093 (0.032)	1.239 (0.022)	0.803 (0.046)	0.755 (0.042)
Agriculture, Forestry, Fishing and Hunting [NAICS 11]	1.252 (0.091)	1.126 (0.045)	1.022 (0.081)	1.036 (0.085)
Arts, Entert, and Recr [NAICS 71]	1.898 (0.058)	1.400 (0.023)	0.792 (0.034)	0.763 (0.021)
Construction [NAICS 23]	1.016 (0.065)	1.010 (0.058)	0.842 (0.040)	0.948 (0.027)
Educational Services [NAICS 61]	2.073 (0.059)	1.636 (0.027)	0.693 (0.021)	0.819 (0.019)
Finance and Insurance [NAICS 52]	0.902 (0.027)	0.951 (0.016)	0.954 (0.032)	1.043 (0.024)
Health Care and Social Assistance [NAICS 62]	1.169 (0.046)	1.164 (0.022)	0.656 (0.018)	0.768 (0.017)
Information [NAICS 51]	1.012 (0.032)	1.074 (0.017)	1.036 (0.031)	0.922 (0.027)
Mng of Companies and Enterprises [NAICS 55]	0.931 (0.056)	0.968 (0.038)	0.873 (0.089)	0.925 (0.050)
Mining, Quar and Oil and Gas Extrac [NAICS 21]	1.055 (0.054)	1.024 (0.027)	1.305 (0.176)	1.079 (0.101)
Prof, Scien, and Tech Services [NAICS 54]	1.011 (0.035)	1.044 (0.020)	0.899 (0.023)	0.911 (0.021)
Real Estate and Rental and Leasing [NAICS 53]	0.920 (0.030)	0.935 (0.020)	0.725 (0.078)	0.786 (0.023)
Retail Trade [NAICS 44/45]	1.530 (0.052)	1.272 (0.025)	0.912 (0.060)	0.783 (0.032)
Trans and Warehousing [NAICS 48/49]	1.022 (0.031)	1.089 (0.018)	0.890 (0.041)	0.917 (0.036)
Utilities [NAICS 22]	0.922 (0.040)	0.999 (0.030)	1.168 (0.082)	1.354 (0.095)
Wholesale Trade [NAICS 42]	1.077 (0.048)	1.044 (0.025)	0.909 (0.038)	0.944 (0.030)
Missing	1.449 (0.138)	1.200 (0.088)	1.174 (0.136)	1.126 (0.119)

Table 18: Firm Outcomes and Lagged Match-State Innovations: Alternative Reference State

	Stock			New Hires		
	Growth	Separation	Survival	Growth	Separation	Survival
inn ₁ (Stepping-stone)	0.107*** (0.001)	−0.146*** (0.001)	0.041*** (0.001)	0.041*** (0.001)	−0.052*** (0.001)	0.016*** (0.000)
inn ₂ (Stable-Stagnant)	0.153*** (0.001)	−0.232*** (0.001)	0.033*** (0.001)	0.127*** (0.001)	−0.193*** (0.001)	0.029*** (0.000)
inn ₃ (Churning)	−0.201*** (0.002)	0.352*** (0.001)	−0.055*** (0.001)	−0.154*** (0.001)	0.137*** (0.001)	−0.032*** (0.000)
inn ₅ (Early Growth)	0.182*** (0.001)	−0.184*** (0.001)	0.034*** (0.000)	0.086*** (0.001)	−0.077*** (0.001)	0.017*** (0.000)
inn ₆ (Maturing Growth)	0.148*** (0.001)	−0.252*** (0.001)	0.029*** (0.000)	0.129*** (0.001)	−0.197*** (0.001)	0.020*** (0.000)
inn ₇ (Stable-Ongoing)	0.145*** (0.001)	−0.260*** (0.001)	0.039*** (0.000)	0.072*** (0.001)	−0.094*** (0.000)	0.008*** (0.000)
Reference state	4	4	4	4	4	4
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N (firm-years)	23,911,695	24,784,527	24,901,908	8,506,274	8,655,303	8,671,221

Notes: This table reports estimates of the relationship between match-state innovations and firms' future activity using an alternative reference state. The first three columns construct match-state innovations using the firm's stock of employees, while the final three columns construct them using the firm's new hires. Within each set of columns, the dependent variables are the DHS symmetric employment growth rate from year $t - 1$ to t , the annualized separation rate, and an indicator for firm survival to year t . State 4 in the reordered classification (state 1 in the original classification) is omitted and constitutes the reference category. All specifications include firm and year fixed effects. For each outcome, firms with only one observation in the corresponding estimation sample are excluded. Standard errors clustered at the firm level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Source: Revelio Labs.*